

3

In this chapter, we will learn about

- X-ray radiographic imaging
- CT scanning
- radionuclide imaging
- health risks of these imaging methods

Medical Imaging Using Ionizing Radiation

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3.1 X-ray radiographic imaging

In Books 3B and 5, we discussed X-rays. An important application of X-rays is taking X-ray images for medical diagnostic purpose (Fig 3.1a). Have you ever taken any X-ray images?

Flipped classroom

Watch a video and answer the questions.

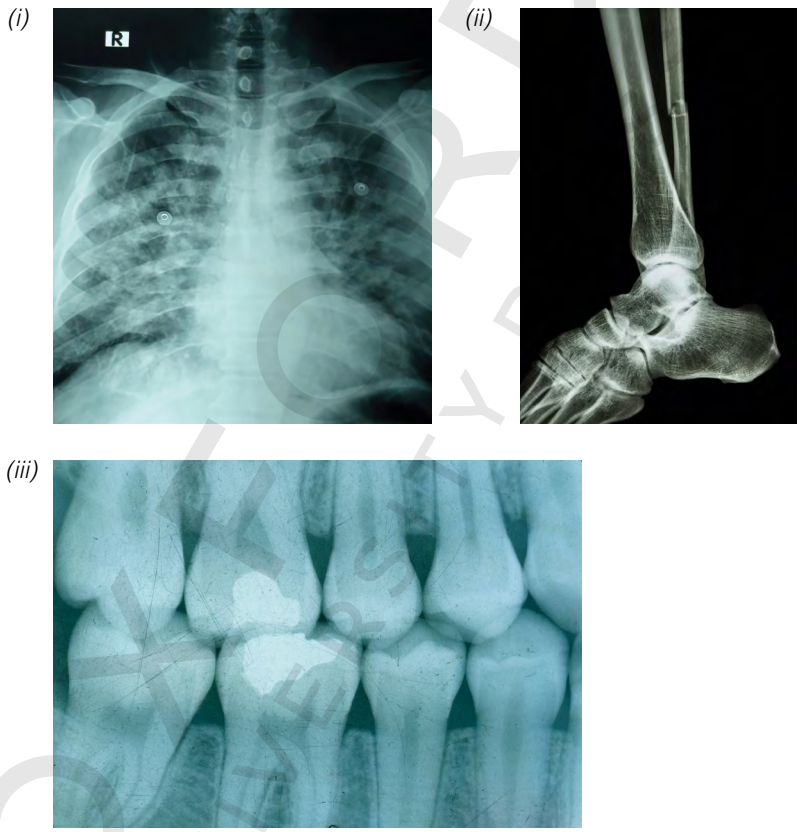


Fig 3.1a *X-ray images of (i) lungs infected by COVID-19, (ii) a leg with broken bone, (iii) teeth with fillings.*

In this section, we shall study how these images are taken.

1 Working principle of X-ray radiographic imaging

The above images are also known as **X-ray radiographic images** or radiographic images. To take such images, the patient is put between an X-ray tube and an X-ray detector (Fig 3.1b on p.117). The X-ray tube emits a short burst of X-rays towards the patient. The X-rays are detected by the detector after passing through the patient's tissues. The signals from the detector are processed to form the image. What happens when the X-rays pass through the patient's tissues? Let us find out the answer below.

X-ray radiographic image X 射線放射攝影圖像

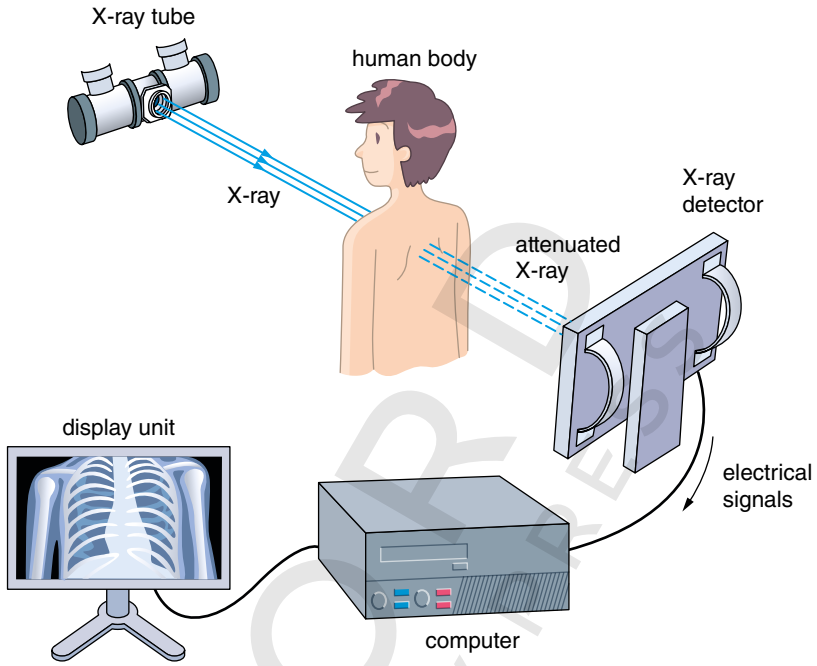


Fig 3.1b *Producing an X-ray image.*

2 Attenuation of X-rays

In Book 5 Chapter 1, we learned about the properties of X-rays. Let us have a review.

Pre-study **X-rays**

X-rays are electromagnetic waves with wavelengths ranging from 0.01 nm to 10 nm, which is shorter than those of ultra-violet radiation (Fig a).

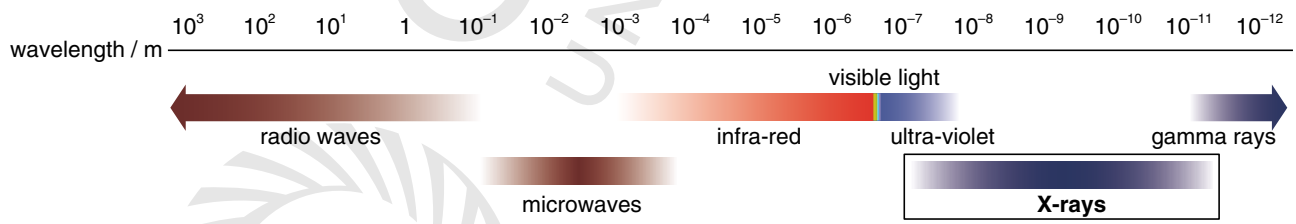


Fig a

X-rays have the following properties:

1 They have high penetrating power.

2 They are ionizing radiation and can damage living cells.

3 They can be produced when fast-moving electrons hit a heavy metal (Fig b).

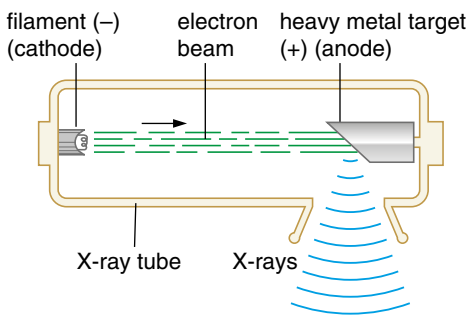


Fig b

DSE goal

Apply $I = I_0 e^{-\mu x}$ to determine the transmitted intensity of an X-ray beam.

Relate the linear attenuation coefficient to half-value thickness.

DSE exam

16(2)Q4.7, 17(2)Q.4(d), 18(2)Q4.7, 19(2)Q4.8, 20(2)Q.4(b), 21(2)Q4.6, 22(2)Q4.7, 23(2)Q4.6

Similar to ultrasound, X-rays lose energy when travelling through a medium. Its intensity is reduced along the distance travelled. In other words, attenuation also occurs to X-rays.

The degree of attenuation depends on two factors: the medium through which the X-ray travels and the distance travelled by the X-ray in the medium. Figure 3.1c shows how the intensity of X-rays changes along the distance travelled in some body tissues. You may now understand why the X-rays passing through different parts of a patient's body would have different changes in intensity (Fig 3.1d).

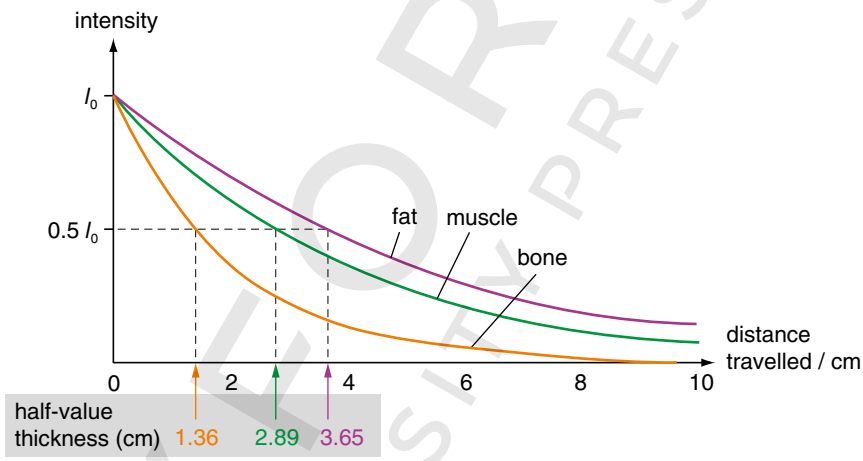


Fig 3.1c *Attenuation of X-rays in some body tissues.*

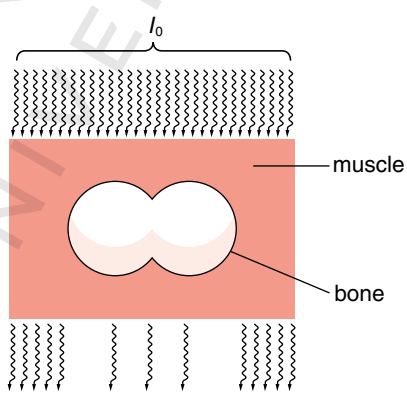


Fig 3.1d *X-rays have a lower intensity after passing through a bone.*

a Half-value thickness

To indicate how a certain material attenuates X-rays, it is useful to know the **thickness of a material that is needed to reduce the intensity of X-ray to half of its original value**. This thickness is called the **half-value thickness**, represented by $x_{½}$ or HVT . It is usually measured in cm. In Figure 3.1c, the half-value thickness for the bone, muscle and fat are 1.36 cm, 2.89 cm and 3.65 cm respectively. In a material with **larger half-value thickness**, the **intensity** of an X-ray beam will **decrease less** for the same distance travelled.

half-value thickness 半值厚度

The following example shows how the **half-value thickness** can be obtained from a graph of intensity against distance travelled.

Example 1 | Half-value thickness

Knowing the half-value thickness of a material is important for radiation protection. With this data, one can find out how thick the protection material is needed to shield radiation.

A patient wears a shield apron to protect her lower abdomen from ionizing radiation when taking a chest X-ray (Fig a). Figure b shows how the intensity of an X-ray beam changes as the beam travels through the material making the apron.



Fig a

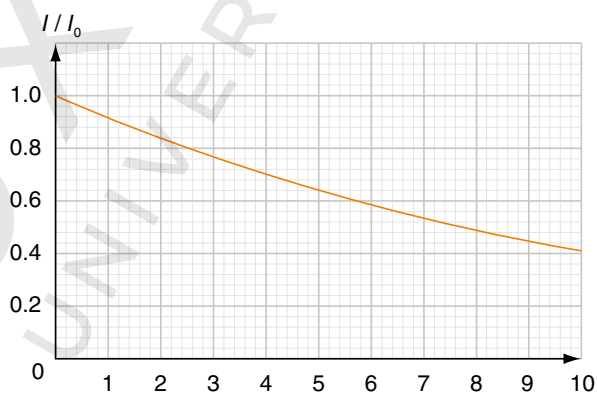


Fig b

(a) Find the half-value thickness $x_{½}$ of the material.

(b) If an X-ray beam of initial intensity I_0 passes through a distance of $2x_{½}$ in the material, what will its intensity I be?

Solution

Tip
Read the value of x corresponding to $0.5I_0$.

(a) 7.8×10^{-3} cm

Cross-link
The skills used to solve these problems are similar to those used to solve problems about half-life in Book 5 Ch 2.

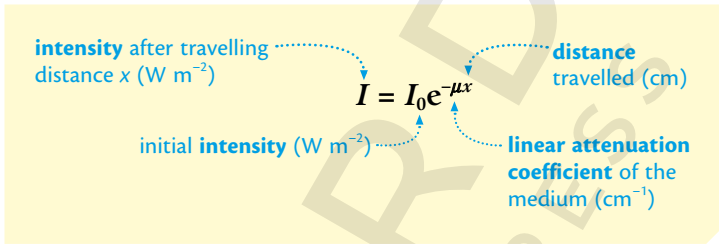
(b) $I = I_0 \times \frac{1}{2} \times \frac{1}{2}$

$= \frac{1}{4} I_0$

b Linear attenuation coefficient

The change in intensity over distance travelled can be described by an exponential decay function. Suppose a beam of X-ray with initial intensity I_0 enters a medium. After travelling a distance x, its intensity I is given by (Fig 3.1e):

✎ The universal nature of the exponential equation in describing natural phenomena should be stressed here.



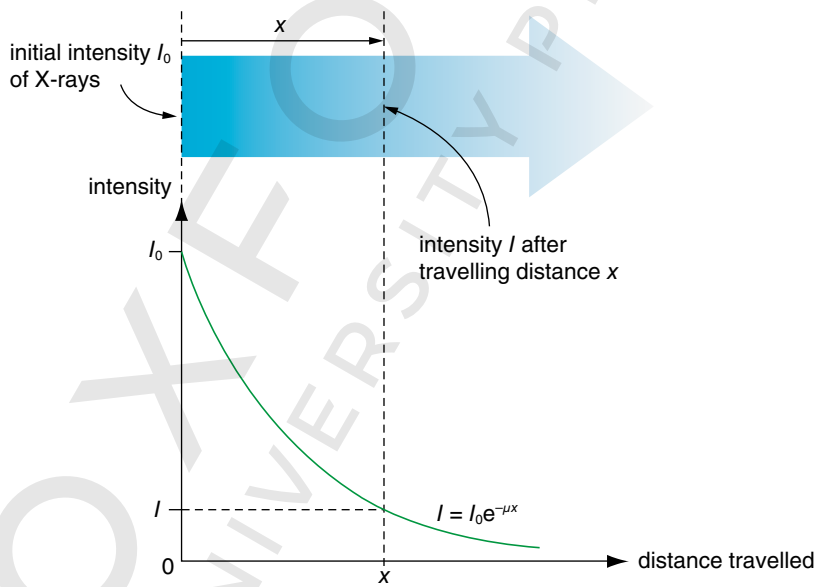


Fig 3.1e *Change in intensity over distance travelled.*

where μ is the **linear attenuation coefficient** of the medium. The unit of μ is cm^{-1} when that of x is cm.

When an X-ray beam has travelled one half-value thickness $x_{½}$, its intensity drops by half from I_0 to $\frac{1}{2}I_0$. Therefore, the relationship between the linear attenuation coefficient μ and the half-value thickness $x_{½}$ can be obtained, which is

When intensity drops by half, we have

$$\frac{1}{2}I_0 = I_0e^{-\mu x_{½}}$$
$$\frac{1}{2} = e^{-\mu x_{½}}$$
$$e^{\mu x_{½}} = 2$$
$$\mu x_{½} = \ln 2$$

$$x_{½} = \frac{\ln 2}{\mu}$$

half-value thickness (cm) — **linear attenuation coefficient** (cm^{-1})

linear attenuation coefficient 線衰減係數

From this relation, we can see that a material with **smaller linear attenuation coefficient** has a larger half-value thickness, which means the **intensity** of an X-ray beam will **decrease more slowly** in the material.

smaller μ ⇔ larger $x_{½}$ ⇔ I decreases more slowly (smaller attenuation)

Table 3.1a shows some typical values of linear attenuation coefficient and half-value thickness.

Medium	μ / cm^{-1}	$x_{½}$ / cm
air	0.000 27	2600
water	0.23	3.0
fat	0.19	3.6
soft tissues	0.23	3.0
blood	0.24	2.9
muscle	0.24	2.9
bone	0.51	1.4

Table 3.1a *Some typical values of linear attenuation coefficient and half-value thickness for X-rays with frequency 1.2 × 10^{19} Hz.*

Note that:

1 **Air** has a very **small linear attenuation coefficient**. It allows almost all X-rays to pass through it.

2 **Bone** has a **relatively large linear attenuation coefficient**. The intensity of X-rays is reduced quickly when passing through it.

3 In general, a medium with higher density has a larger linear attenuation coefficient.

density of medium ↑ ⇒ μ ↑

The linear attenuation coefficient of a medium also depends on the frequency or energy of the X-rays used. If the X-ray beam has a higher frequency or a higher energy, the linear attenuation coefficients of the media it passes through will become smaller and hence the X-rays beam will have a larger penetrating power.

frequency/energy of X-ray ↑
⇒ μ ↓
⇒ penetrating power of X-ray ↑

Example 2 | Linear attenuation coefficient

The figure below shows an X-ray image of a human hand. It is given that the linear attenuation coefficients of muscle and bone are 0.24 cm^{-1} and 0.51 cm^{-1} respectively.



(a) Find the percentage of incoming intensity of X-rays that passes through
   (i) muscle of thickness of 1.5 cm (fleshly part of palm),
   (ii) a bone of thickness of 1 cm (the finger).

(b) How will the result in (a)(ii) change if X-rays of higher energy are employed? Explain your answer.

Solution

(a) (i) By $I = I_0 e^{-\mu x}$,

$$\frac{I}{I_0} = e^{-\mu x} = e^{-0.24 \times 1.5} = 0.698$$

$$\text{Percentage of intensity} = \frac{I}{I_0} \times 100\% = 0.698 \times 100\% = 69.8\%$$

(ii) By $I = I_0 e^{-\mu x}$,

$$\frac{I}{I_0} = e^{-\mu x} = e^{-0.51 \times 1} = 0.600$$

$$\text{Percentage of intensity} = \frac{I}{I_0} \times 100\% = 0.600 \times 100\% = 60.0\%$$

(b) X-rays of higher energy have a higher penetrating power. Therefore, a larger percentage of X-rays will pass through the bone, i.e. the result in (a)(ii) will increase.

Cross-link

The skills used to solve these problems are similar to those used to solve problems about decay constant in Book 5 Ch 2.

Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 Which of the following statements about X-ray are correct? Put a tick in the corresponding box(es).

- ☐ X-ray is an electromagnetic wave.
- ☐ X-ray is an ionizing radiation.
- ☐ X-ray has a high penetrating power.
- ☐ X-ray can be produced by bombarding a heavy metal target with high-energy electrons.
- ☐ X-ray has higher frequency than gamma rays.

2 An X-ray beam is attenuated when passing through a layer of material. Which of the following changes would cause the attenuation to become higher? Put a tick in the corresponding box(es).

- ☐ A thicker layer is passed through.
- ☐ A layer of material with a higher linear attenuation coefficient is passed through.
- ☐ A layer of material with a higher density is passed through.
- ☐ An X-ray beam of higher frequency is used.





Learn by practice 1

1 Find the linear attenuation coefficient of a material with a half-value thickness of 4 cm.

2 Find the half-value thickness of a material with a linear attenuation coefficient of 0.2 cm^{-1}.

3 Find the percentage of the intensity of an X-ray beam remained after travelling in water for 10 cm. It is given that the linear attenuation coefficient of water is 0.23 cm^{-1}.

4 An X-ray room is surrounded by thick concrete walls. It is given that the linear attenuation coefficient of concrete is 0.82 cm^{-1}. If the walls are built to reduce the intensity of the X-rays to 1% of its original value, what is the thickness of the concrete walls?

5 An X-ray beam has an intensity of 36.8 W cm^{-2} after travelling 4 cm through a medium with a linear attenuation coefficient of 0.25 cm^{-1}. What is the initial intensity of the X-ray beam?

6 An X-ray beam **loses** 80% of its initial intensity after travelling 2 cm in a medium. Find the linear attenuation coefficient of the medium.

C X-rays passing through multiple media

When an X-ray beam of initial intensity I_0 passes through media 1 and 2 one after another, it is attenuated by medium 1 and then by medium 2.

Intensity after passing through medium 1 = $I_0 \times e^{-\mu_1 x_1}$

Intensity after passing through medium 2 = $(I_0 \times e^{-\mu_1 x_1}) \times e^{-\mu_2 x_2}$

$= I_0 \times e^{-\mu_1 x_1} \times e^{-\mu_2 x_2}$

$= I_0 \times e^{-\mu_1 x_1 - \mu_2 x_2}$

$= I_0 \times e^{-(\mu_1 x_1 + \mu_2 x_2)}$

In general, for an X-ray beam passing through multiple media (Fig 3.1f):

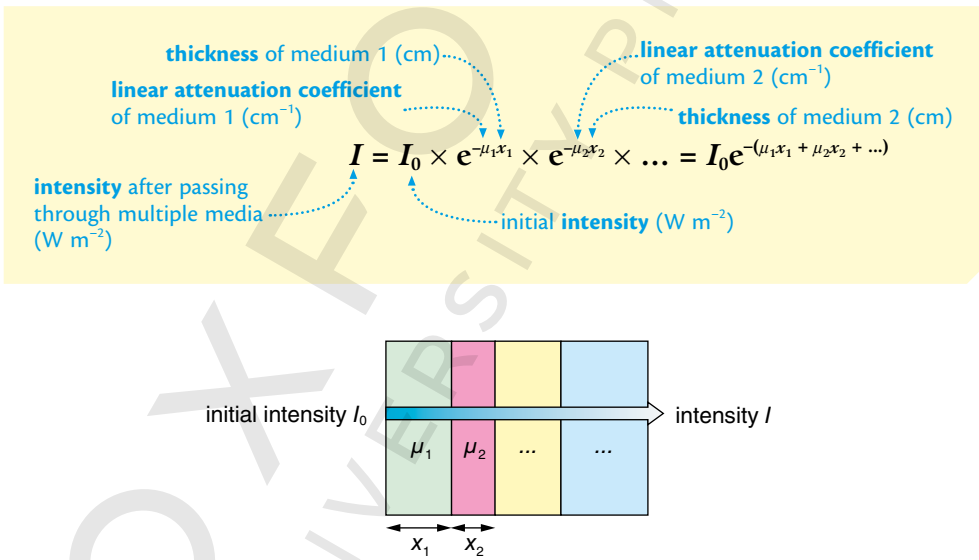


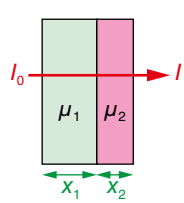
Fig 3.1f Change in intensity when passing through multiple media.

Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 An X-ray beam passes through layers 1 and 2 of different materials as shown. The thicknesses of layers 1 and 2 are x_1 and x_2, and the linear attenuation coefficients are μ_1 and μ_2 respectively. Find the percentage of the intensity of the X-ray beam remained after passing through the layers according to the values given in the following table.



	x_1 / cm	x_2 / cm	μ_1 / cm^{-1}	μ_2 / cm^{-1}
(a)	5	2	0.5	0.24
(b)	3.2	4.6	0.23	0.0003

☆ HKDSE 2014 Paper 2 Q4.6

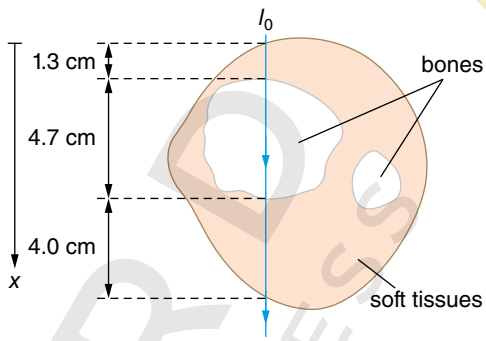
→ This video demonstrates how this question is solved.

Video

DSE practice 1 | X-rays transmitted through multiple media

An X-ray beam of intensity I_0 passes through part of a leg as shown. After travelling a distance x, the intensity of the beam becomes I.

Assume that bones have a linear attenuation coefficient of 0.51 cm^{-1} and the soft tissues have a linear attenuation coefficient of 0.23 cm^{-1}.



(a) The graph of $\ln\left(\frac{I}{I_0}\right)$ against x for x = 0 to 1.3 cm is a straight line. State the slope of the graph and sketch the graph. (2 marks)

(b) Find $\frac{I}{I_0}$ when x = 6 cm. (3 marks)

Solution

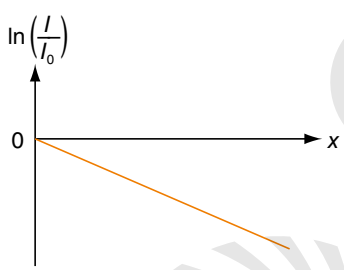
(a) By $I = I_0 e^{-\mu x}$,

$$\frac{I}{I_0} = e^{-\mu x}$$

$$\ln\left(\frac{I}{I_0}\right) = -\mu x$$

The equation is rearranged into the form $y = mx + c$ with $\ln\left(\frac{I}{I_0}\right)$ as y and x as x. Therefore,
- It is a straight line.
- Slope $m = -\mu$
- It passes through origin as y-intercept c = 0.

Slope = $-\mu$ = −0.23 cm^{-1} 1A



(Graph line passing through origin with negative slope) 1A

(b) For x = 6 cm, the X-ray beam has passed through 1.3 cm soft tissues and 4.7 cm bone.

$I = I_0 \times e^{-\mu_1 x_1} \times e^{-\mu_2 x_2}$ 1M

Common mistake ⚠
Students may not know how to apply $I = I_0 e^{-\mu x}$ for multiple times.

$$\frac{I}{I_0} = e^{-\mu_1 x_1 - \mu_2 x_2}$$

$$= e^{-(\mu_1 x_1 + \mu_2 x_2)}$$

$$= e^{-(0.23 \times 1.3 + 0.51 \times 4.7)}$$ 1M

$$= 0.0675$$ 1A

▶ Revision 3 Q13 (p.172)

DSE goal

Realize a radiographic image as a map of attenuation of X-ray beam after passing body tissue.

DSE exam

13(2)Q.4(b)(i), 20(2)Q4.8

3 Formation and interpretation of X-ray images

a Formation of X-ray images

To produce an X-ray image, a patient is positioned between an X-ray tube and an X-ray detector. The detector records the intensity of X-rays transmitted through the body (Fig 3.1g).

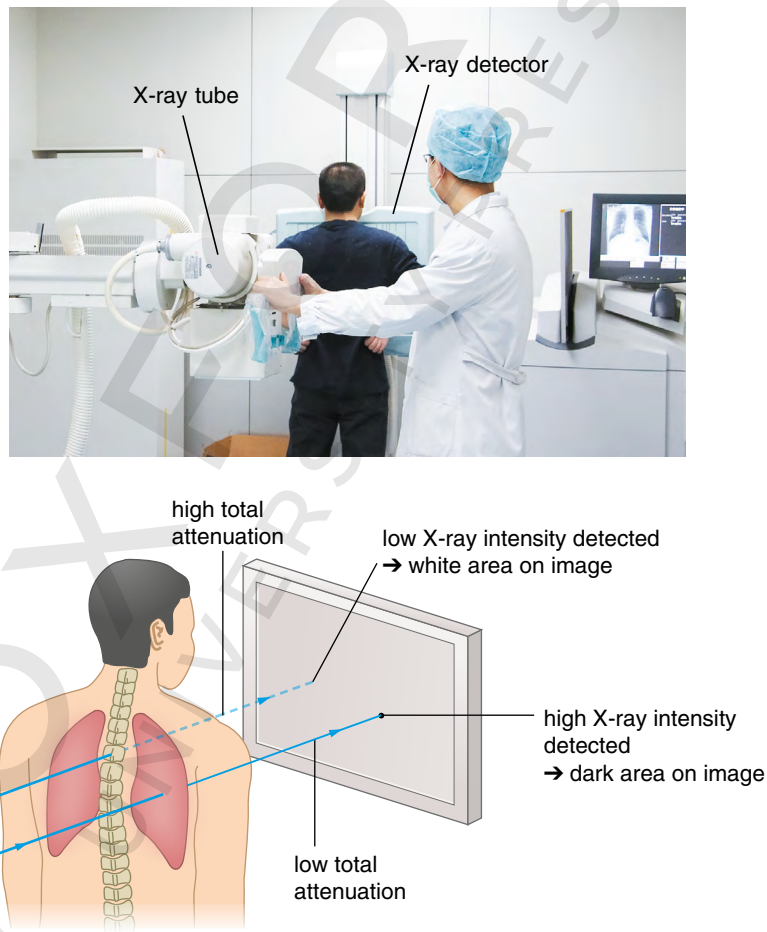


Fig 3.1g *A patient positioned between the X-ray tube and the X-ray detector.*

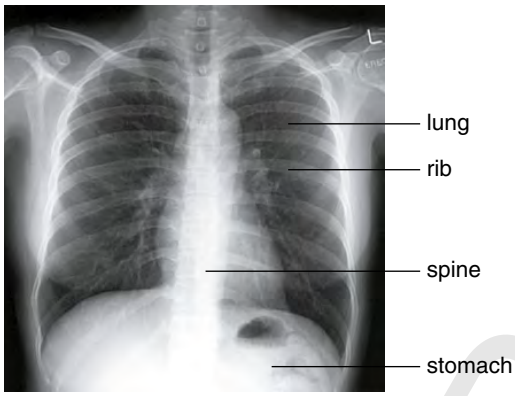
✎ Unlike detectors of ultrasound imaging, which detect ultrasound **reflected**, those of X-ray radiography detect X-rays **transmitted**.

Most medical procedures nowadays use digital detectors to produce X-ray images. However, photographic films are still used in some medical centres.

▶ The data collected from the detector are processed and then transferred to a display unit. The shade of grey displayed depends on the X-ray intensity detected, and is therefore related to the **total attenuation** at the particular body parts the X-ray beams have passed through. A region is darker if the X-ray intensity detected is higher.

higher X-ray intensity detected ↔ darker area on the image ↔ lower total attenuation at body parts that the beam has passed through

Figure 3.1h shows an X-ray image of the chest.



Total attenuation depends on the attenuation coefficient of each organ as well as its thickness. the X-ray beams have passed through. Therefore,

- thick bones (like the spine) appear white;
- thin bones (like the ribs) appear grey;
- thick soft structures (stomach region) appear white;
- lungs containing mostly air appear black.

Fig 3.1h *Different shades of grey displayed on an X-ray image of the chest.*

An X-ray image can be viewed as a map of attenuation of X-ray beams. Beams passing through body parts with low total attenuation produce black regions, while beams passing through body parts with high total attenuation produce white regions.

Know more

From array of numbers to image

Digital images are formed by putting a large number of pixels together. When the X-rays reach the detector, an array of numbers describing the intensities of X-rays detected at different positions is obtained (Fig a). Each of these numbers decides the darkness of a pixel. These pixels then combine to form an X-ray image (Fig b).

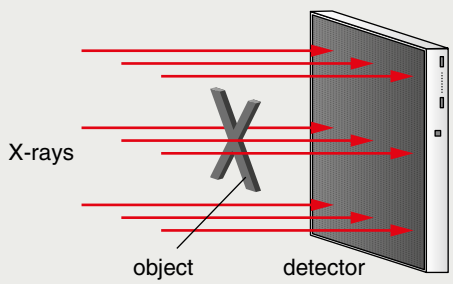
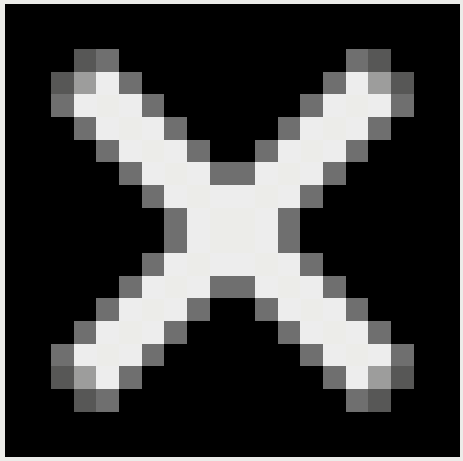


Fig a

10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10
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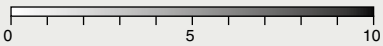


Fig b

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 An X-ray image is a map of (reflection / attenuation) of X-ray beams.

2 Thick bony structures has a (low / high) linear attenuation coefficient and allow a (small / large) amount of X-rays to pass through. These structures usually appear (white / black) on X-ray images.

3 Which of the following body tissues allows the largest amount of X-rays to pass through? Assume that they are of the same thickness.

A Fat B Lungs
C Bone D Heart

Writing practice

Question

Explain the appearance of the lungs, i.e. region *A*, on the X-ray image (Fig a). (3 marks)

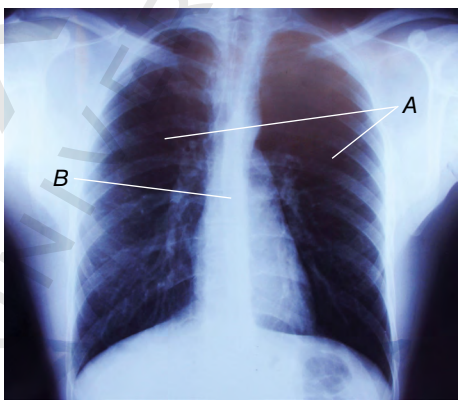


Fig a

Answer

As air inside the lungs has a very low linear attenuation coefficient, 1A

X-rays passing through the lungs have a very low attenuation 1A

and produce a black region on the image. 1A

Practise yourself

1 Explain the appearance of the spine, i.e. region *B*, on the X-ray image (Fig a). (3 marks)

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b Interpretation of X-ray images

Conversely, the different shades of grey on an X-ray image can give us hints on what the X-ray beams have passed through. Figure 3.1i shows two examples.

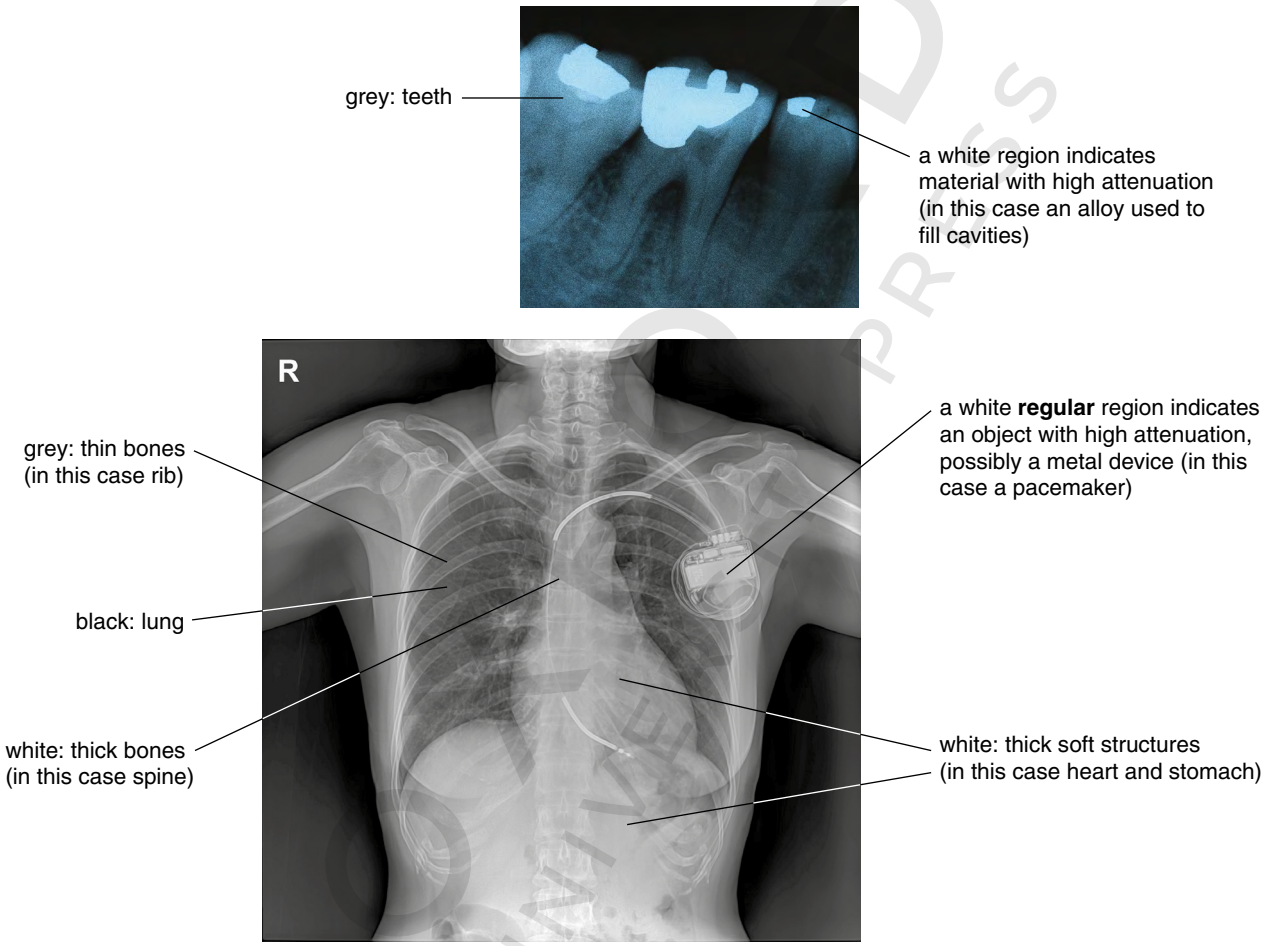


Fig 3.1i *Interpreting X-ray images.*

DSE goal

Explain the use of artificial contrast media such as barium meal in radiographic imaging.

DSE exam

14(2)Q4.7, 23(2)Q4.7

4 Artificial contrast media

In Figure 3.1i, we can see that it is very easy to distinguish the heart and the lung, but hard to distinguish the stomach and the bowel. The **difference in attenuation**, which is called **contrast**, accounts for this. The higher the contrast, the easier it is for us to distinguish the structures.

Some organs have very low contrast with their surroundings, but we can identify them by putting some **artificial contrast medium** inside them. An artificial contrast medium has a high linear attenuation coefficient. If an organ is filled with it, less X-rays can pass through and the organ will appear lighter than normal in an X-ray image.

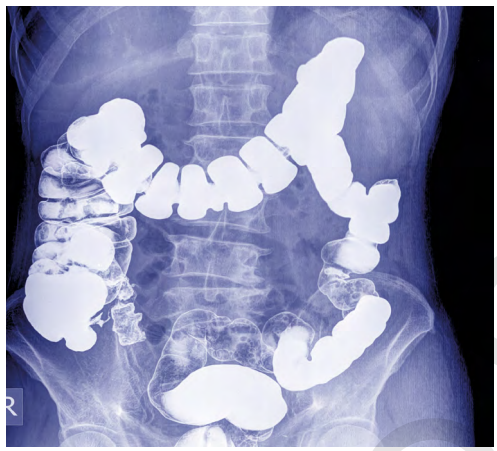
artificial contrast medium 人工顯影劑

Artificial contrast medium is swallowed or injected before imaging is performed. Usually, barium is used for imaging of the digestive tract and iodine is used for imaging of blood vessels (Fig 3.1j).

Barium sulphate can be used as a contrast medium for patients to swallow. This is therefore known as a barium meal.

✎ Remind Ss that barium sulphate is not radioactive.

✎ The contrast medium (iodine) can be injected directly into the uphill artery. We can take X-ray images of the area as blood carries the contrast to the downhill arteries. We call this an angiogram (angio = artery; gram = picture). We do not inject barium as it will clot up the artery and block off circulation.



(i) Barium study of the bowel.



(ii) Iodine study of the coronary artery.

Fig 3.1j *Examples of using artificial contrast media.*

An artificial contrast medium has a high linear attenuation coefficient and can increase the contrast between a body part and its surroundings.

Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* An organ filled with an artificial contrast medium shows up as white region in an X-ray image. (T/F)

2 What is the function of artificial contrast media in X-ray radiographic imaging?

A Increase the X-ray attenuation of the target organ

B Decrease the X-ray attenuation of the target organ

C Increase the X-ray attenuation of tissues surrounding the target organ

D Decrease the X-ray attenuation of tissues surrounding the target organ

5 Practical cases of X-ray radiographic imaging

When doctors look at X-ray images of patients, they know what the image should look like, and they look for abnormalities in the images, such as fractured bones, collapsed spine, enlarged heart, fluid in lungs, *tumours*, loss of cartilage between bones, etc. See the following cases.

coronary artery 冠狀動脈　　tomour 腫瘤

Case 1

patient's X-ray image



X-ray image of a normal leg



Bones can be easily identified due to the high contrast between the bone and the soft tissues.

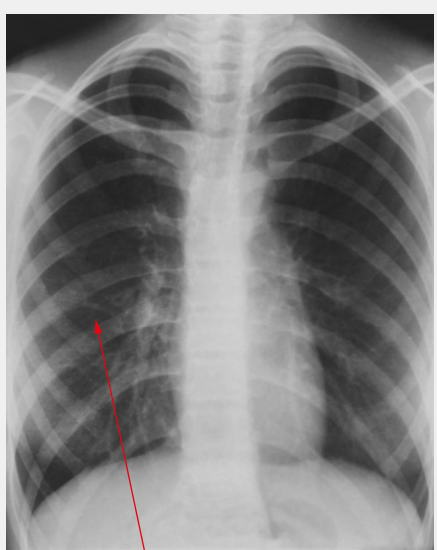
The shin bone is fractured.

Case 2

patient's X-ray image



X-ray image of a normal chest

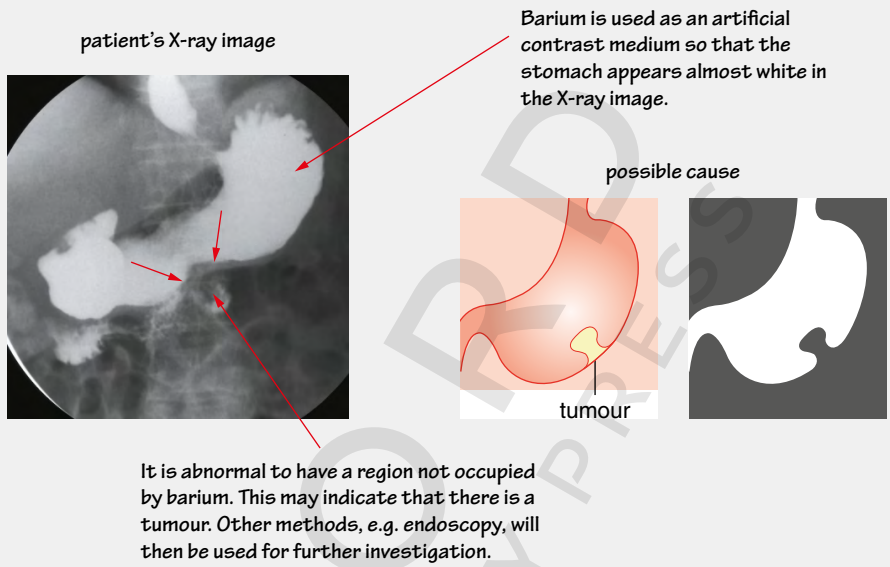


Ground glass opacities are observed in the lung area of a COVID patient.

Lung tissues are filled with air and appear almost black in the X-ray image.



Case 3



DSE goal

Discuss the advantages and disadvantages of radiographic imaging in diagnosis.

6 Advantages and disadvantages of X-ray radiographic imaging in diagnosis

Using X-ray radiographic imaging in diagnosis has several advantages and disadvantages (Table 3.1b).

Actually, we can perform continuous X-ray imaging to obtain dynamic images but the radiation dose is high. ▶

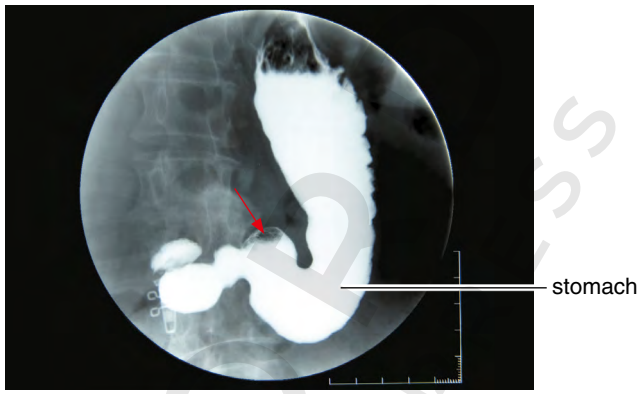
Advantages	Disadvantages
• It is a **fast** method of medical imaging. Broken bones and lung diseases can quickly be diagnosed. • It is **cheap and widely available**. This is important for screening diseases in a large population, e.g. taking chest X-ray images for suspected carriers of *tuberculosis* or other lung infections.	• There are low but existent **risks of ionizing radiation**. • Different structures are sometimes hard to distinguish due to **overlapping of structures**. • **The images produced are usually static**. The movement of structures cannot be seen. • The **contrast between soft tissues is limited**, though we can improve this by using an artificial contrast medium. • There is a **potential risk of an *allergic* reaction to the artificial contrast medium**.

Table 3.1b *Advantages and disadvantages of X-ray radiographic imaging.*

tuberculosis 肺結核　　allergic 過敏

Example 3 | X-ray image

A patient takes an artificial contrast medium before taking an X-ray image of his stomach. The X-ray image is shown below.



(a) Explain the function of the artificial contrast medium in taking the image shown in the figure.

(b) An abnormality indicated by the red arrow is found in the X-ray image, endoscopy is then used to examine the stomach.

(i) Explain why X-ray imaging is suitable to be used at the beginning for screening for abnormalities.

(ii) State one disadvantage of using endoscopy over X-ray imaging.

Solution

(a) The artificial contrast medium fills the stomach and increases the attenuation of X-rays in passing through the stomach significantly. Therefore, the stomach appears lighter than normal in the image and becomes easier to be distinguished from the surrounding tissues.

(b) (i) X-ray imaging is fast and cheap. It does not require any preparation before imaging.

(ii) Endoscopy is an invasive procedure. The patient may feel uncomfortable during the procedure.

STSE

Importance of X-ray radiographic imaging in screening lung disease

Chest X-ray imaging is commonly used as a first-line investigation for suspected lung infections, especially in *epidemic* situations. For example, if a person is infected with tuberculosis (TB), some white regions will show up on their chest X-ray images.

TB was a common infectious disease in Hong Kong in the 1960s. In more recent lung infection outbreaks, such as corona virus disease 2019 (COVID-19), severe acute respiratory syndrome (SARS) and *avian influenza*, chest X-ray imaging was also used in diagnosis of suspected cases.

✎ Other ways to detect TB include the skin test and blood test, but neither is as quick as a chest X-ray.

epidemic 傳染病　　avian influenza 禽流感

Exercise 3.1

1 An X-ray beam has an initial intensity of 200 W cm^{-2}. Its intensity is attenuated to 50 W cm^{-2} after travelling 6 cm in a medium. Find the half-value thickness of the medium.

A 1.2 cm B 1.5 cm
C 2 cm D 3 cm

***2** Which of the following does the half-value thickness of a medium for X-rays depend on?

(1) The density of medium
(2) The frequency of the X-ray travelling in the medium
(3) The distance travelled by the X-ray in the medium

A (1) only B (1) and (2) only
C (2) and (3) only D (1), (2) and (3)

3 State two advantages and two disadvantages of X-ray radiographic imaging.

4 (a) State the meaning of half-value thickness of a material for X-rays.

(b) A material has a half-value thickness of 8 cm.

(i) Find its linear attenuation coefficient.

(ii) If an X-ray beam enters the material, find the percentage of its intensity remained after travelling 24 cm in the material.

***5** An X-ray beam has an initial intensity of 100 W cm^{-2}. Its intensity is attenuated to 20 W cm^{-2} after travelling 8 cm in a medium.

(a) Find the linear attenuation coefficient of the medium.

(b) Hence, find the half-value thickness of the medium.

***6** An X-ray beam travels 2 cm in soft tissues and then 5 cm in bone. The linear attenuation coefficients of soft tissue and bone are 0.23 cm^{-1} and 0.51 cm^{-1} respectively.

(a) In which medium, soft tissues or bone, does the X-ray beam lose energy more quickly?

(b) Find the percentage of the intensity of the X-ray beam remained.

***7** The graph below shows how the intensity of an X-ray beam changes as it passes through a distance x in a material (Fig a).



Fig a

(a) Estimate the half-value thickness of the material.

(b) Hence, estimate the linear attenuation coefficient of the material.

(c) Find the thickness of this material needed to reduce the intensity of the X-rays to 10% of its original value.

***8** Figures b and c are two X-ray images showing the abdomen of a normal person and a person with colon cancer respectively. The colons of the two persons are highlighted by filling them with an artificial contrast medium before taking the images.



Fig b



Fig c

(a) Explain why the artificial contrast medium can highlight the colons from other body structures.

(b) Locate the tumour in Figure c and explain how the tumour is highlighted in the image.

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*9 Figures d and e show X-ray images of a normal chest and a patient's chest respectively.

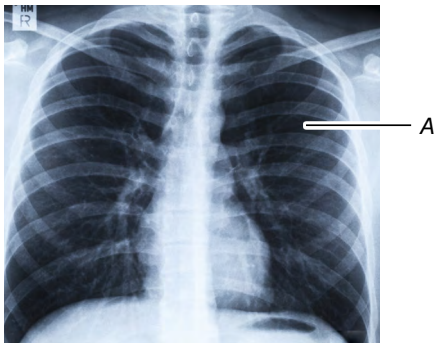


Fig d



Fig e

(a) Explain why region *A* in the image of a normal chest is almost black.

(b) Hence, explain how the image of the patient's chest shows the possibility of lung disease.

*10 Figure f shows an X-ray image of a metal pipe.

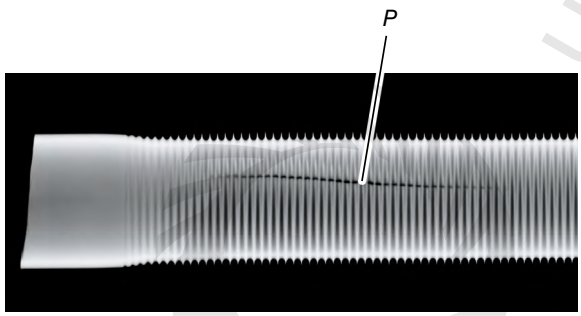


Fig f

(a) Briefly describe how this image is produced.

(b) Region *P* in the image appears darker than other parts of the pipe. What can you say about the thickness of metal in this region?

*11 Osteoporosis is a disease caused by the loss of minerals in bones. As minerals are lost, the density of the bone decreases. This weakens the strength of bone and makes the bone be broken easily. Figures g and h are two X-ray images of the hands of two persons. One shows a normal hand and the other one shows a hand with osteoporosis.



Fig g



Fig h

Which figure is more likely to be the image showing the hand with osteoporosis? Explain your answer.

3.2 CT scanning

The following images are also produced by using X-rays but they look very different from radiographic images. What are these images? How are they produced? What can they tell us? Let us find out the answers in this section.

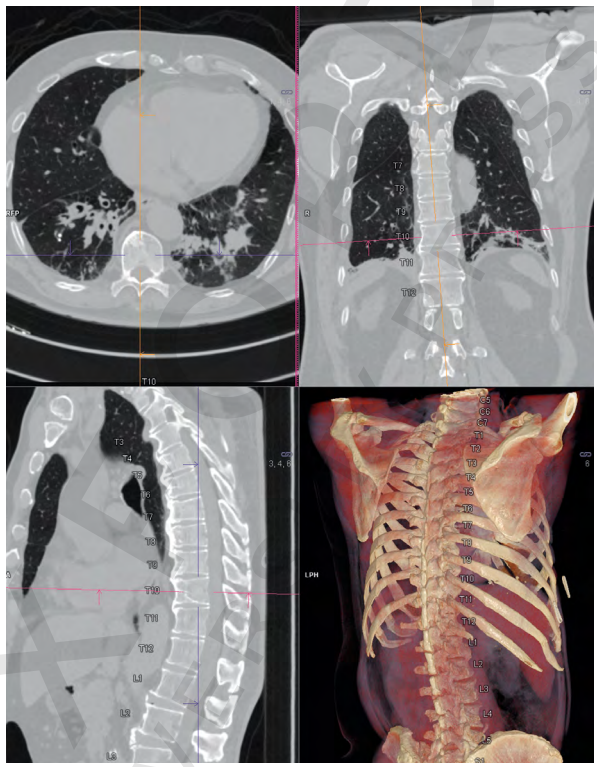


Fig 3.2a *Some images produced by using X-rays.*

The above images are **computed tomographic images** (**CT images**). As you can see from Figure 3.2a, CT images can show different cross-sections and even three-dimensional views of the human body.

As its name tells us, CT images are produced using a technique called **computed tomography** (**CT**). It is an advanced application of X-rays to obtain a *tomograph*, i.e. an image of a **cross-section**, of the human body (Fig 3.2b). This allows doctors to examine body tissues in greater detail than what simple X-ray radiographic imaging can do.

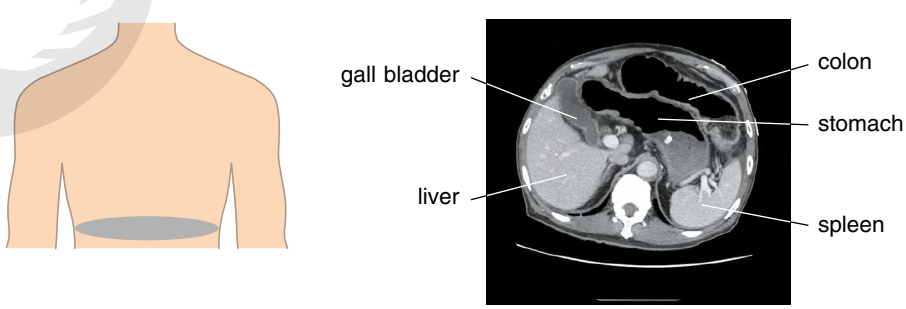


Fig 3.2b *A CT image showing the cross-section of the abdomen.*

computed tomographic image (CT image) 電腦斷層影像
computed tomography (CT) 電腦斷層造影　　tomograph 斷層造影圖

DSE goal

Describe how a CT scanner works.

Realize a CT image as a map of attenuation coefficients of body tissues.

DSE exam

13(2)Q.4(b), 18(2)Q4.8

1 Working principle of CT scanning

To produce a CT image, the patient is moved through a CT scanner (Fig 3.2c). The scanner contains an X-ray tube and a row of X-ray detectors (Fig 3.2d). During a CT scan, both the tube and the detector rotate together around the patient (Fig 3.2e).



Fig 3.2c *A patient is moved through a CT scanner.*

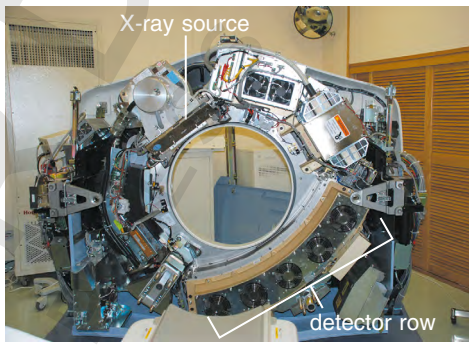


Fig 3.2d *Internal structure of a CT scanner.*

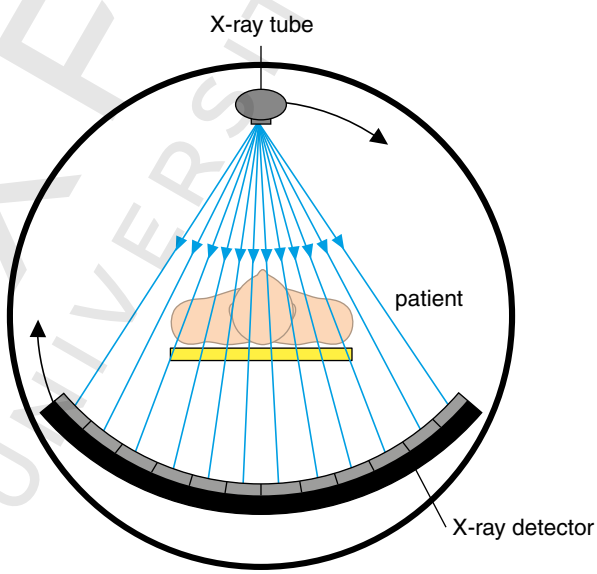


Fig 3.2e *An X-ray tube and an X-ray detector rotate in a circle.*

a Production of a CT image of a slice of body

Multiple tomographs are usually obtained during a CT scan. Let us first focus on how the tomograph of a slice of body is obtained.

① During the scan, a fan beam of X-rays is directed to pass through a slice of the patient's body. Like in the case of radiographic imaging, different X-ray beams are attenuated by different amounts when travelling inside the body. The X-rays are then picked up by the row of detectors and their recorded intensities are processed to obtain an **attenuation profile** (Fig 3.2f on p.138).

attenuation profile 衰減輪廓

② As the X-ray tube and the detector rotate around the patient, a number of attenuation profiles are obtained.

Hence this imaging method is called computed tomography.

③ Data from the attenuation profiles are fed to the computer, which uses a method called **back projection algorithm** to reconstruct a map of attenuation coefficients of the cross-section of the body.

④ The map is then filled different shades of grey to form a CT image. A region is filled darker if the attenuation coefficient there is smaller.

Simulation

→ This simulation demonstrates how a CT image is obtained. It is available on OUP web in PC version (.exe) only.

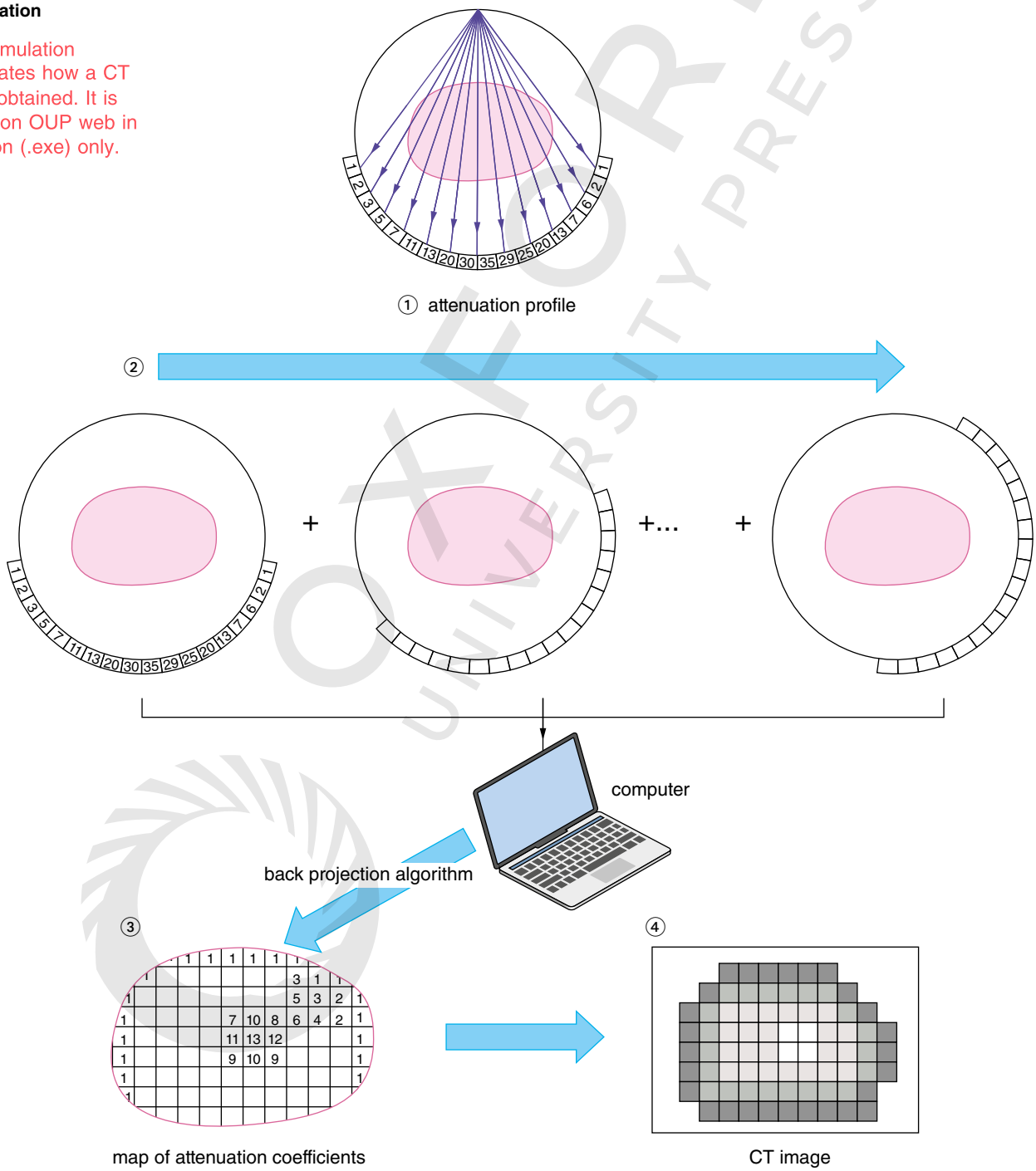


Fig 3.2f *A simplified example showing how a CT image is produced with attenuation profiles.*

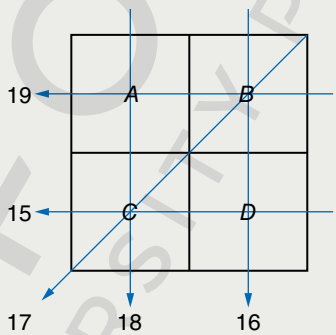
back projection algorithm 反投影算法

To produce a CT image, the X-ray tube and detector rotate round a slice of a patient's body to obtain multiple attenuation profiles. The profiles are used to produce a map of attenuation coefficients of different body tissues by back projection algorithm. This map of attenuation coefficients is used to generate the CT image.

Know more

Calculating attenuation coefficients by back projection algorithm

Let us use a 2 × 2 'slice' to show how back projection algorithm works. Suppose the total attenuations along three directions through the slice are recorded as shown.



We can set up 5 equations:

$A + B = 19$, $C + D = 15$, $A + C = 18$, $B + D = 16$, $B + C = 17$

$\therefore (A + B) + (A + C) = 19 + 18$

$2A + (B + C) = 37$

$2A + 17 = 37$

$A = 10$

From this we can solve that $A = 10$, $B = 9$, $C = 8$ and $D = 7$. This gives the attenuation coefficients of different parts in the slice.

The resolution in real CT scans is 512 × 512 or even higher. The calculation is much more complicated and is performed by computer.

DSE exam
15(2)Q4.6

b Matrix size and resolution of CT image

An arrangement of numbers in rows and columns is called a *matrix*. Therefore, a CT image can also be regarded as a matrix of attenuation coefficients. A conventional CT image is a matrix of 512 rows and 512 columns, i.e. its matrix size is 512 × 512. The larger the matrix size, the higher the resolution of a CT image is. The detectors in a modern CT scanner are aligned much more densely than in earlier ones, and hence the resolution of CT images has been greatly improved over the years.

matrix 矩陣

Example 4 | Matrix size

The CT image in Figure a is the first patient image of the brain scanned on the prototype scanner in 1971 and has a matrix size of 80 × 80. Figure b is a modern CT image of the brain.

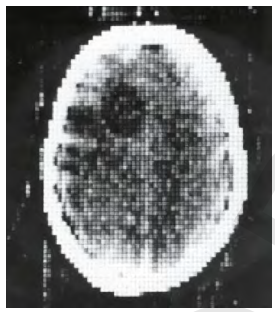


Fig a

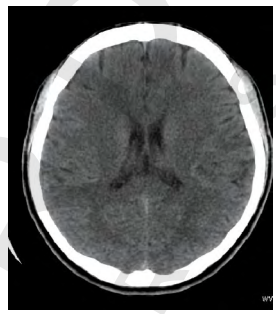


Fig b

If the image size of the modern CT image is 32 cm × 32 cm and the size of each pixel of the image is 0.0977 mm^2, what is its matrix size?

Solution

Length of a pixel = $\sqrt{0.0977}$ = 0.3126 mm

Number of pixels along one dimension = $\frac{\text{length of image}}{\text{length of pixel}} = \frac{32 \times 10}{0.3126} = 1024$

∴ Its matrix size is 1024 × 1024.

With larger matrix size, it is now possible to identify fine details of the brain. You can see that the edges of the brain in the modern CT image are much smoother. ▶

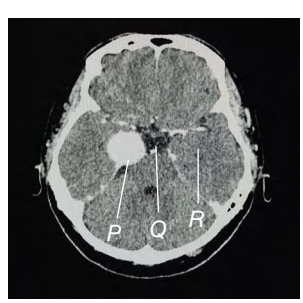
Quick check 5

Google Forms Microsoft Forms → E-version in OUP Exercise Platform is also available.

1 A CT image is a map of
- A intensity of X-rays passing through the body.
- B intensity of X-rays reflected from the body.
- C attenuation coefficients of different parts of the body.

2 To construct a CT image, we need to obtain
- A only one attenuation profile.
- B many attenuation profiles taken from different angles.
- C many attenuation profiles taken at different times.

3 A CT image of brain is shown below. Which part (*P*, *Q* or *R*) has the lowest attenuation coefficient?



4 The (smaller / larger) the matrix size, the higher the resolution of a CT image is.

5 The size of an old CT image is 32 cm × 32 cm. If its matrix size is 80 × 80, find the size of each pixel of the image.

DSE goal

Realize the image reconstruction process of CT scanning.

c Production of CT images of multiple slices of body

During a scan, the patient is slowly moved through the scanner on a bed. By doing so, CT images of multiple slices of the body would be produced, allowing us to view cross-sections at different levels through the body (Fig 3.2g). These images together provide us with very detailed information about the condition inside the body.

Watch a video to see how a CT scan is performed practically. ▶

Video

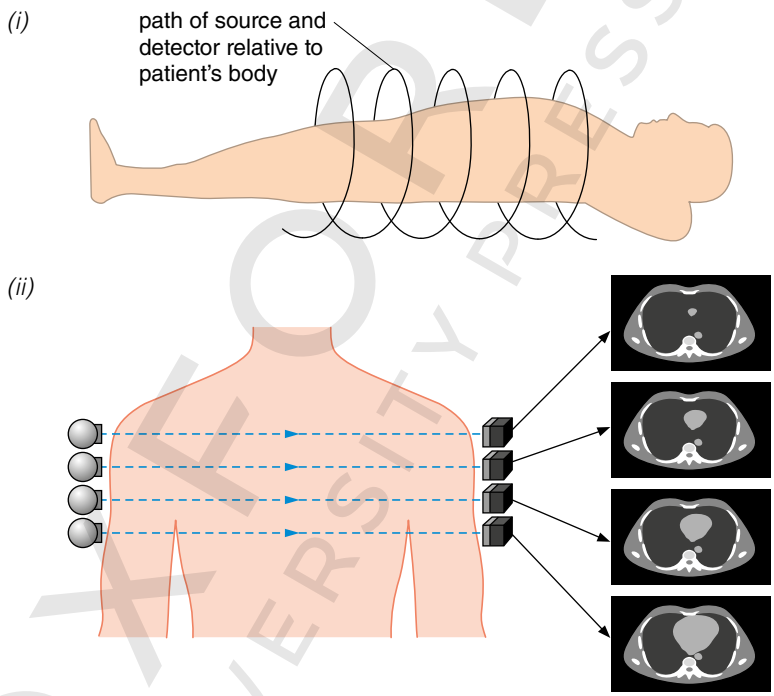


Fig 3.2g *The X-ray tube and detector (i) move in a spiral path relative to the patient's body and (ii) produce cross-sections at different levels along the body.*

All of this post-processing (multi-planar reformation, 3-D reconstruction) is now standard in evaluating a patient's CT scan. This allows a more detailed assessment of disease (e.g. a fracture in terms of the 3-D relationship between fractured fragments). To the patients, they also notice that the images look clearer and are easier to understand. CT has helped demystify medicine in this sense.

d Multi-planar reformation and three-dimensional reconstruction

The CT scanner provides us with attenuation coefficient maps of the tissues at different levels along the body. By stacking these maps, we effectively obtain the attenuation coefficient map of a three-dimensional block (Fig 3.2h).

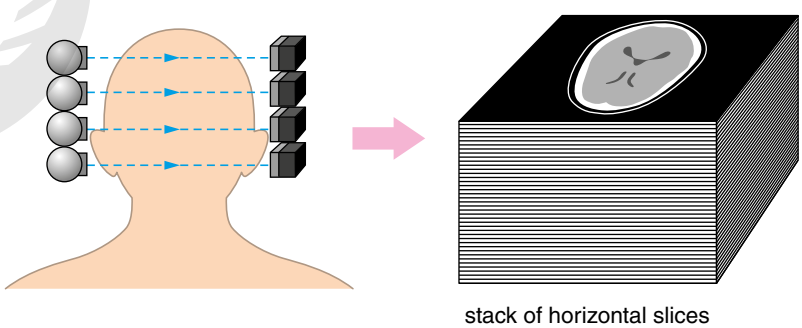


Fig 3.2h *Producing an attenuation coefficient map of a 3-D block.*

We may digitally cut through (reformat) the 3-D attenuation coefficient map and look at cross-sections from any direction. This is called **multi-planar reformation**. Figure 3.2i shows some image slices of the head viewed from different directions.

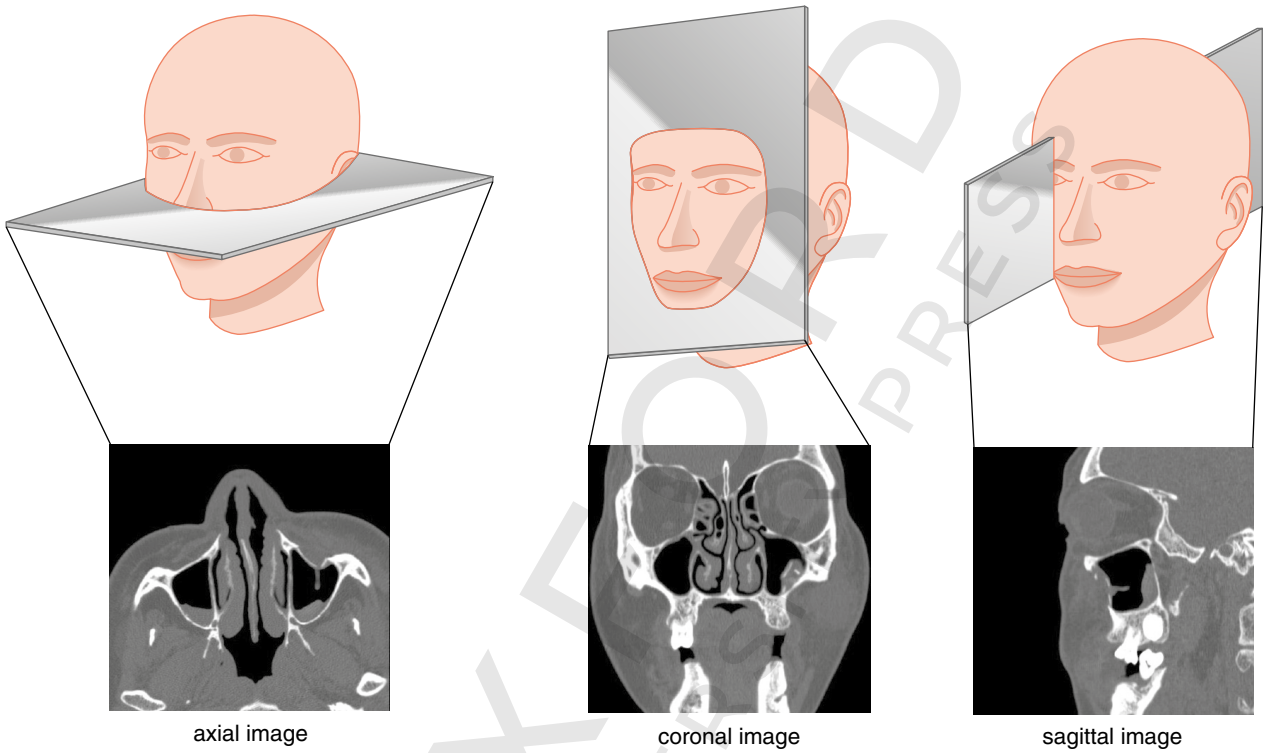


Fig 3.2i *Cross-sectional images of the head in different directions.*

We may also extract parts of the 3-D attenuation coefficient map and fill with different colours to construct 3-D models of body structures (Fig 3.2j & 3.2k). This is called **three-dimensional reconstruction**.

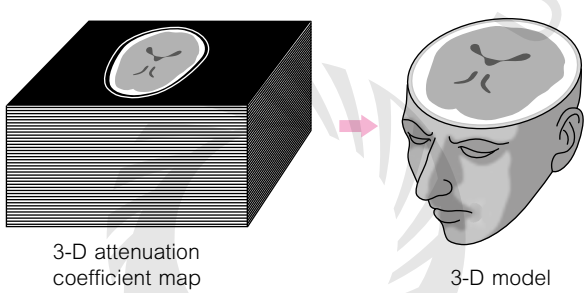


Fig 3.2j *Constructing 3-D model of the head from the 3-D attenuation coefficient map.*



Fig 3.2k *Constructing 3-D model of the skull by extracting the white part of the map.*

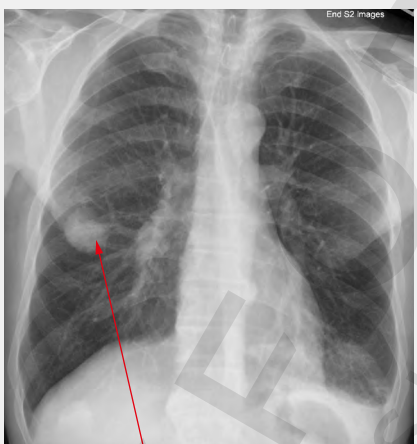
All these digital post-processed images are very helpful in showing the relationship between structures, particularly when doctors are trying to find out how close a disease is away from important structures.

2 Practical cases of CT scanning

Let us now look at some CT images to see how they help doctors identify abnormalities.

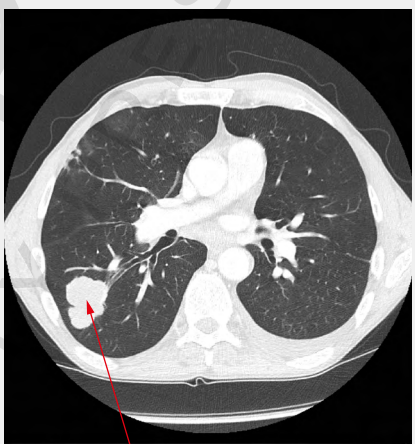
Case 1

X-ray image



X-ray radiographic imaging is typically used as first-line imaging for bone fracture and chest disease. In this case, an abnormal white region is found in the lung.

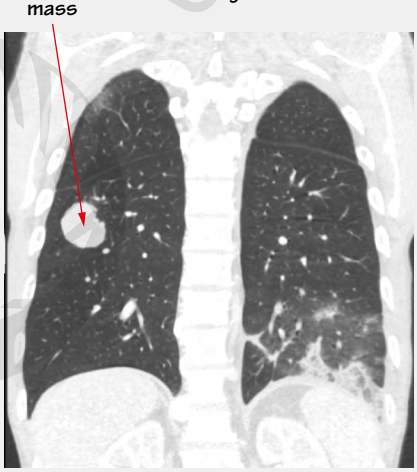
axial CT image



CT scanning is then used for further investigation. A mass is located and it is later confirmed to be a tumour.

coronal reformatted CT image

mass

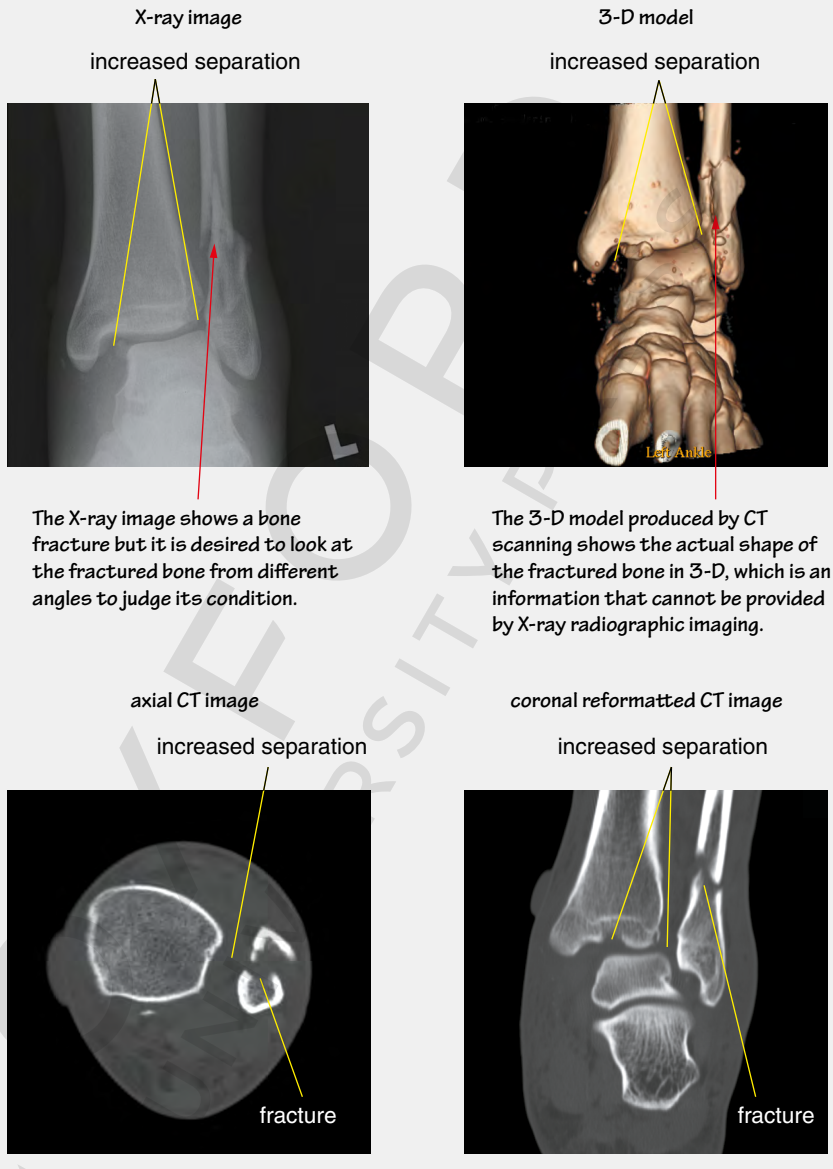


sagittal reformatted CT image

mass



Case 2



DSE goal

Compare CT images with X-ray radiographic images.

DSE exam

13(2)Q.4(b)(iii), 19(2)Q4.5

3 Comparing CT scan and X-ray radiographic imaging

A basic CT image is a 2-D map of **linear attenuation coefficients of tissues** in a cross-section of the body (Fig 3.2l on p.145), while an X-ray image is a 2-D map showing the **total attenuation of X-rays after passing through the body** from one direction (Fig 3.2m on p.145). An X-ray image is the projection of a 3-D structure onto a 2-D plane.

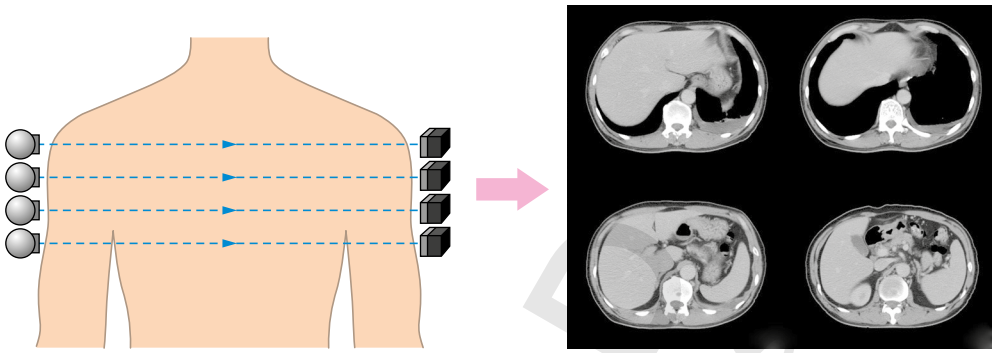


Fig 3.2l CT images.

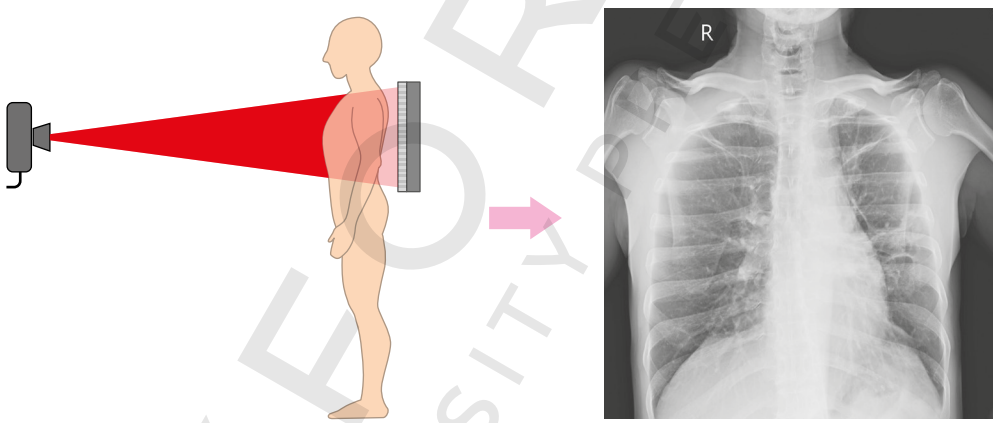


Fig 3.2m X-ray radiographic image.

Therefore, CT scanning provides much more information about the body than X-ray radiographic imaging. The following are some of its advantages:

1 No structure overlapping

Structures that are overlapped in an X-ray radiographic image are clearly separated in a CT image.

2 Accurate location and shape

The 3-D model generated from CT scanning can show the accurate location and shape of structures in 3-D space (Fig 3.2n), while an X-ray radiographic image cannot show the depth and thickness of a structure in the body.

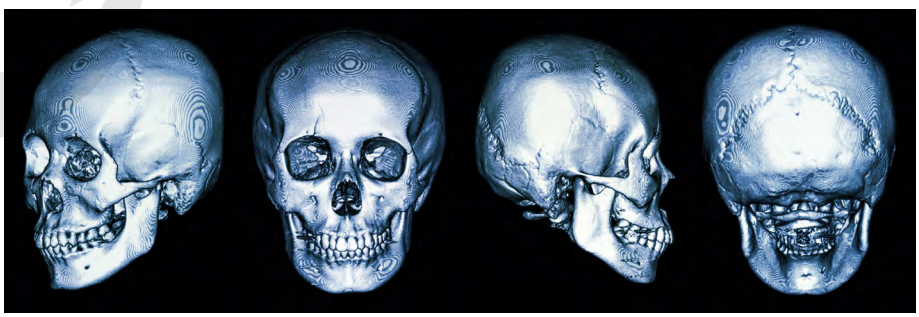


Fig 3.2n CT scanning allow us to view the skull from different directions in one go.

However, CT scanning also has its disadvantages when compared to X-ray radiographic imaging. They include:

1 Higher cost and less widely available

CT scanning equipment is more expensive and the cost of producing a CT image is much higher. Therefore, CT scanning is not as widely available as X-ray radiographic imaging.

2 Longer imaging time

A CT scan needs to take a lot of image slices to produce reformats and generate 3-D models while X-ray radiographic imaging does not. Therefore, the time taken for a CT examination (including scanning and reformatting) is longer.

3 Higher radiation exposure

Both imaging methods make use of X-rays. As mentioned, a CT scan requires taking multiple image slices, so a patient who receives a CT scan is exposed to more ionizing radiation.

Table 3.2a summarizes the comparison.

Aspect	CT scanning	X-ray radiographic imaging
Cost	☹ higher	🙂 lower
Availability	☹ not as widely available	🙂 widely available
Time taken for imaging	☹ longer	🙂 shorter
Exposure to ionizing radiation	☹ more	🙂 less
Overlapping of body structures	🙂 no overlapping	☹ may overlap
Generation of 3-D image	🙂 capable	☹ incapable

Fig 3.2a *Comparison between CT scanning and X-ray radiographic imaging.*

Google Forms Microsoft Forms

Quick check 6

→ E-version in OUP Exercise Platform is also available.

1 Which of the following statements about CT imaging is/are correct? Put a tick in the corresponding box(es).

- ☐ It makes use of X-rays.
- ☐ It can produce three-dimensional images of body structures.
- ☐ It takes longer time to perform than X-ray radiographic imaging.
- ☐ In a CT image, we can clearly distinguish structures that are overlapped in an X-ray radiographic image.

Exercise 3.2

★**1** During a CT scan, X-rays are directed to pass through a body part and arrive at a detector as shown in Figure a.

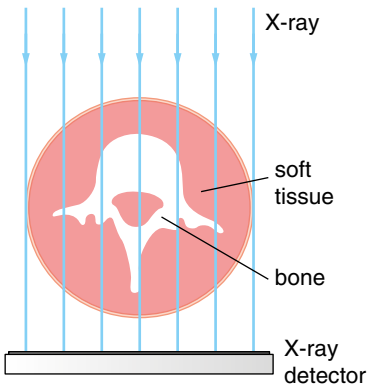
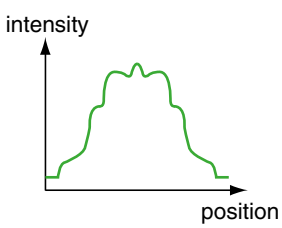


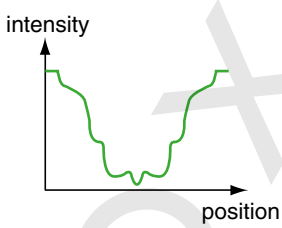
Fig a

Which of the following graph shows the most possible intensity of X-rays detected at different positions along the detector?

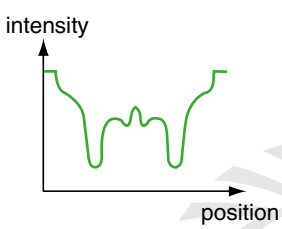
A



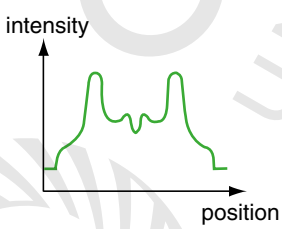
B



C



D



★**2** Which of the following statements about CT scanning and X-ray radiographic imaging is/are correct?

(1) Both of them use ionizing radiation.

(2) Both of them produce images based on the attenuation of X-rays.

(3) Both of them show bones clearly without using artificial contrast medium.

A (1) only B (1) and (2) only

C (2) and (3) only D (1), (2) and (3)

3 A 32 cm × 32 cm image with a pixel size of 0.0977 mm^2 is produced in a CT scan. How many pixels are included on each side of the image?

4 State two advantages and two disadvantages of CT scanning when compared to X-ray radiographic imaging.

★**5** Figure b shows the interior of a CT scanner.

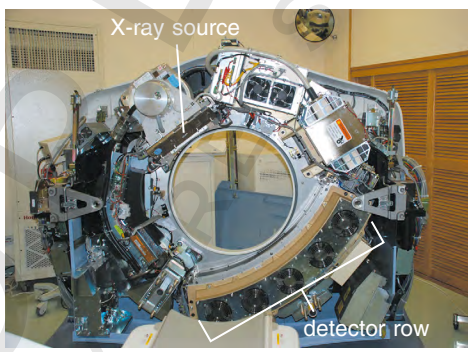


Fig b

Describe how a tomograph of a slice of body is produced in CT scanning. Include the following words in your answer: **X-ray tube**, **X-ray detector**, **attenuation profile**, **back projection algorithm**.

★**6** Figure c shows a post-processed image in a CT scan. It makes use of a technique called the multi-planar reformation.



Fig c

(a) Briefly describe how this image is produced.

(b) A doctor would like to find out if there are blood clots inside the heart. Why is this image preferred over a chest X-ray image as shown in Figure d?

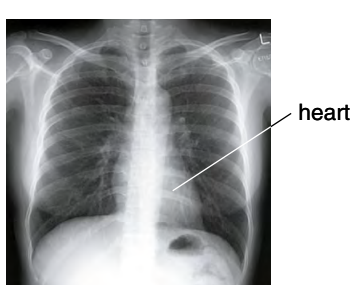


Fig d

3.3 Radionuclide imaging

Can you imagine taking a radioactive source into our body? Does it sound dangerous? Actually this is done in the process of a kind of medical imaging called **radionuclide imaging**, which produces low resolution images (Fig 3.3a). We shall learn more about how this can be done and why such low resolution images are useful.



Fig 3.3a A radionuclide image of the liver.

DSE goal

Describe the use of radioisotopes as tracers for diagnosis.

Realize a radionuclide image obtained by a gamma camera as a map of radioisotopes distribution in a body.

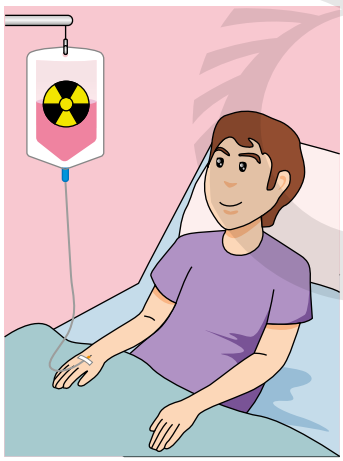
DSE exam

16(2)Q.4(a), 20(2)Q4.6

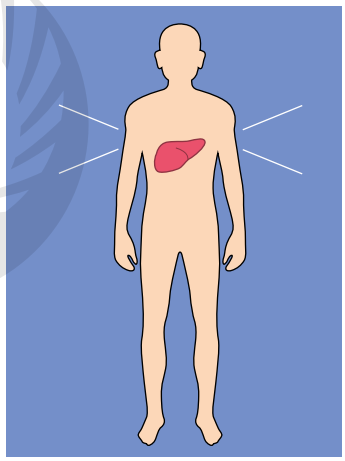
1 Working principle of radionuclide imaging

Figure 3.3b shows the procedures of radionuclide imaging.

① A small amount of radionuclide tracers which emit γ radiation is injected into the patient's body.



② The type of tracer is chosen such that they will accumulate in the target organ or structure to be imaged.



③ The gamma camera detects the γ radiation and then transmits data to a computer to produce an image.

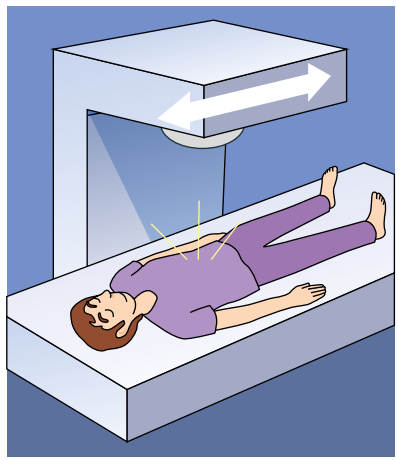


Fig 3.3b Procedures of radionuclide imaging.

radionuclide imaging 放射性核素造影術

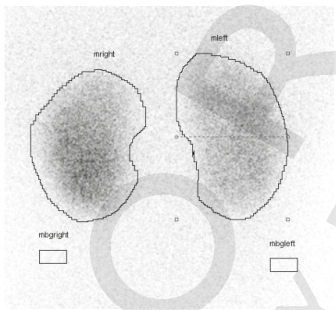
Since the **tracers** are injected into the body, radionuclide imaging is an invasive imaging method.

> **Cross-link**
> In Book E2, we learned about the particle nature of electromagnetic radiation. In fact, each black dot on a radionuclide image corresponds to one γ photon. If more photons are collected, there are more black dots and the corresponding region looks darker.

All tracers used in radionuclide imaging are γ sources. A gamma camera detects γ radiation (Fig 3.3c) and produces an image showing the **distribution of the tracers in the body**. A darker colour indicates a higher concentration of tracers. When the tracers accumulate in the target organ (or structure), a dark region is shown on the image (Fig 3.3d), indicating the position and shape of the organ.



Fig 3.3c *A gamma camera is positioned over the patient when producing images.*



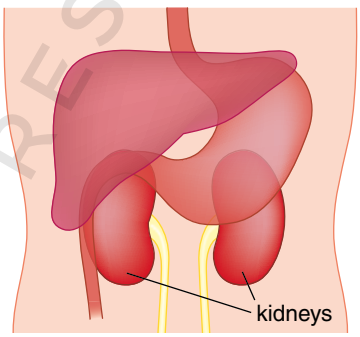


Fig 3.3d *Dark regions indicate where the tracers accumulate in.*

Video

→ This video shows a gamma camera at work.

Note that it takes some time for the tracers to travel around the body. When we observe the tracers at different times, we can see that they are distributed over different parts of the body (Fig 3.3e).

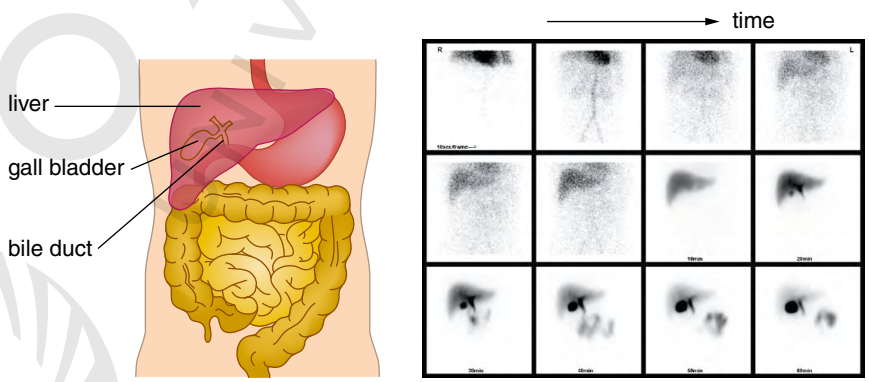


Fig 3.3e *The radionuclides are taken up by the liver and then excreted into the bile duct. Therefore, we see the liver first, then the bile duct and the gall bladder.*

In radionuclide imaging, radionuclide tracers are injected into the patient's body and accumulate in the target organ. The gamma camera detects the γ radiation emitted from the tracers. The data is then processed by a computer to produce a radionuclide image, which is a map of the distribution of the radionuclides in the patient's body.

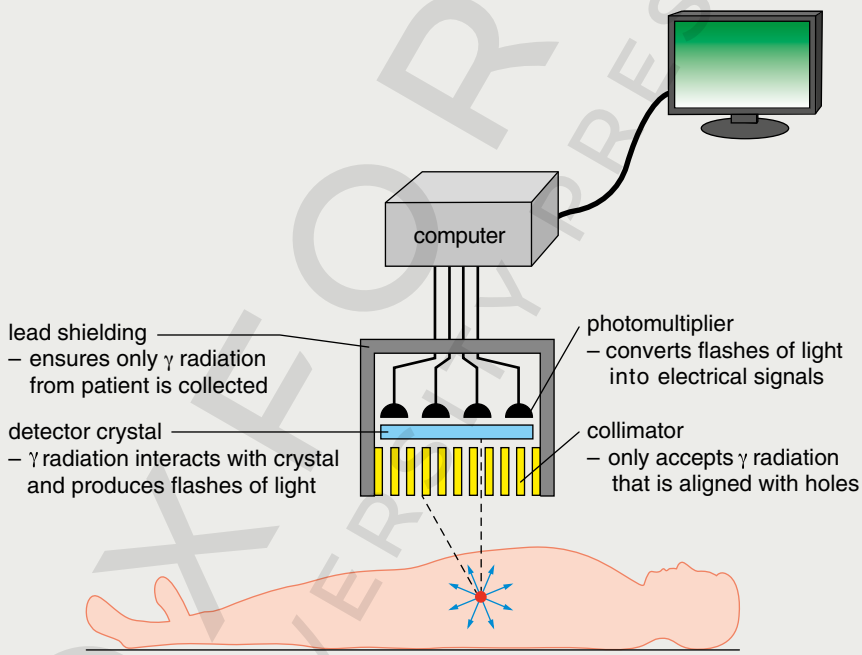
tracer 示蹤物

Know more

Structure of a gamma camera

The figure below shows the structure of a gamma camera. The collimator is a lead plate with holes and it only accepts γ radiation which is perpendicular to the plate. It also has a lead shield to block γ radiations that is not coming from the patient. The incoming γ radiation is picked up by a crystal and converted to visible light, which is in turn converted to electrical signals.

Since the radiation emitted by the tracers is very weak, it takes some time for the camera to collect enough radiation to form an image. In some cases, it requires more than 20 minutes to take a single image.



Components of a gamma camera.

Google Forms Microsoft Forms

Quick check 7

→ E-version in OUP Exercise Platform is also available.

1 Radionuclide images of a target organ are produced by using a gamma camera to detect γ radiation

A emitted from the organ.

B reflected by the organ.

C transmitted through the organ.

2 Arrange the following steps in the correct order to show the general process of radionuclide imaging.

a Radionuclides accumulate at the target organ.

b Radionuclides are injected into the patient's body.

c An image of the target organ is produced.

d Radiation emitted by the radionuclides is collected by a gamma camera.

2 Use of radionuclides as tracers

a Delivery of tracers

Radionuclide tracers are usually delivered into a patient's body by injecting into the blood stream (Fig 3.3f). The tracers would then be circulated to the target body part by the blood stream. In some cases, the tracers can also be delivered by inhalation (e.g. imaging for the lungs), by direct injection into an organ (e.g. imaging for lymph nodes) or by taking orally.



Fig 3.3f *Syringe is equipped with a lead shield to protect the staff from the radiations during the injection of a radionuclide tracer.*

Cross-link

Recall from Book 5 Ch 2 that isotopes are nuclides of the same element that have different numbers of neutrons.

After being injected into the body, the tracers may accumulate in an organ through two ways. Some tracers are radioisotopes of elements that are **naturally absorbed** by the target organs, while some are **combined with other substances which are normally taken up** by the target organs.

Know more

Examples of tracer absorption

Let us see some examples of tracer absorption in more detail.

▸ p.178

Know more

Common tracers

Some common tracers for different organs and structures are listed below.

	Target organ or structure	Radionuclide tracer
Natural absorption	thyroid gland	iodine-123
	heart	thallium-201
	lung	xenon-133
Combined with substances	bone	technetium-99m-dicarboxypropane diphosphonate
	brain	technetium-99m-exametazime
	salivary gland	technetium-99m-pertechnetate
	kidney	technetium-99m-dimercaptosuccinic acid
	liver	technetium-99m with sulphur colloid

DSE goal

Define biological half-life and apply it to solve related problems.

DSE exam

14(2)Q4.8, 16(2)Q.4(b), 17(2)Q4.7, 19(2)Q4.7, 23(2)Q4.8

b Half-lives of tracers

After the radionuclide is introduced into a patient's body, its activity decreases. The reasons are two-fold: the radionuclide itself decays, and it gradually leaves the patient's body.

i Physical half-life

Recall that radionuclides decay exponentially with time and the time it takes for half of the radionuclide to decay is its half-life, or more specifically, its **physical half-life**.

The physical half-life of a radionuclide is the time taken for half of the nuclei in a sample to decay.

ii Biological half-life

For example, the biological half-life of water in a human is about 7 to 10 days. Therefore in one week, about half of the water molecules in our body are replaced.

When we take in some radionuclides into our bodies, our bodies gradually expel the substance by usual metabolic process and excretion such as urination. Therefore, the **biological half-life** of the substance should also be considered when using tracers.

The biological half-life of a substance is the time taken for half of the substance to be removed from the body by biological processes.

iii Effective half-life

Now, we see that the amount of radionuclide remaining in a patient's body depends on its physical half-life (t_p) and biological half-life (t_b). To describe their combined effect, we introduce the quantity **effective half-life** (t_e).

The effective half-life is the time taken for the activity of a radionuclide in a human body to fall to half of its original value.

In this equation, the units for t_e, t_p and t_b must be the same.

$$\frac{1}{t_e} = \frac{1}{t_p} + \frac{1}{t_b}$$

effective half-life (any unit of time)
physical half-life (any unit of time)
biological half-life (any unit of time)

physical half-life 物理半衰期　biological half-life 生物半衰期　effective half-life 有效半衰期

Example 6 Effective half-life

The physical and biological half-lives of iodine-131 are 8 days and 138 days respectively.

(a) What is the effective half-life t_e of iodine-131?

(b) A patient receives iodine-131 for radionuclide imaging. Find the time needed for the amount of radionuclides remaining in his body to drop to 10%.

Solution

(a) By $\frac{1}{t_e} = \frac{1}{t_p} + \frac{1}{t_b}$,

$$t_e = \left(\frac{1}{t_p} + \frac{1}{t_b}\right)^{-1} = \left(\frac{1}{8} + \frac{1}{138}\right)^{-1} = 7.56 \text{ days}$$

(b) Decay constant $k = \frac{\ln 2}{t_e}$

By $N = N_0 e^{-kt}$,

$$t = \frac{1}{-k} \ln \frac{N}{N_0}$$

$$= \frac{1}{-\frac{\ln 2}{t_e}} \ln 0.1 = \frac{7.56}{-\ln 2} \ln 0.1 = 25.1 \text{ days}$$

$\therefore$ The time needed is 25.1 days.

Google Forms Microsoft Forms

Quick check 8

→ E-version in OUP Exercise Platform is also available.

1 In most radionuclide imaging processes, radionuclides are carried to the target organ by

A medical instruments.

B internal mechanisms of the body.

C gamma emission.

2 Complete the following table about the half-lives of different radionuclides.

	Radionuclide	Physical half-life	Biological half-life	Effective half-life
(a)	*A*	87.4 days	90 days	
(b)	*B*	8 days		7.6 days
(c)	*C*		24 hours	4.8 hours

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DSE goal

Identify the characteristics of radionuclides such as technetium-99m used for diagnosis.

DSE exam

12(2)Q4.8, 15(2)Q4.8, 20(2)Q4.7

C Choosing radionuclides as tracers

A radionuclide suitable for medical imaging should have the following characteristics.

1 It should **emit γ radiation only**.

Among α, β and γ radiation, γ radiation has the weakest ionizing power, and thus causes the least amount of cellular damage inside our body. Moreover, having a high penetrating power, γ radiation emerging from inside the patient can easily be detected externally.

2 The γ radiation emitted should be of **suitable energy**.

The energy of γ radiation should be low enough to make it safe for use, yet high enough to make it be readily detected.

3 It should have a **suitable physical half-life and biological half-life**.

Both half-lives should be long enough to maintain a high activity for imaging, but short enough to reduce total radiation exposure. Another reason for a short physical half-life is to present minimal problems for disposal. In general, a radionuclide with physical half-life and biological half-life of a few hours is optimal.

4 It should be **non-toxic** and have **no *pharmacological* effect** on the patient.

5 It can be **combined with a wide range of substances to form tracers** for imaging different organs.

6 It should **decay to a stable nuclide**.

7 It should be **readily available** and **inexpensive**.

✎ α particles cannot penetrate the skin. β particles can be stopped by a layer of clothing. γ rays are never fully stopped but several metres of concrete or a few centimetres of lead can substantially absorb the rays.

✎ Remind Ss the radioactive sources in radionuclide imaging lies within the patient's body.

Other commonly used radionuclides include iodine-123 and xenon-133. ▶

Technetium-99m (Tc-99m) is one of the most commonly used radionuclides. It has most characteristics that are desirable for medical imaging (Fig 3.3g).

$^{99m}_{43}Tc \rightarrow {}^{99}_{43}Tc + \gamma$ ▶

✎ Technetium is a silver-grey metal. It has no stable isotope and is therefore extremely rare to occur naturally. It plays no natural roles and is not normally found in the human body. Most of the technetium is produced in nuclear reactors. The 'm' in technetium-99m stands for metastable.

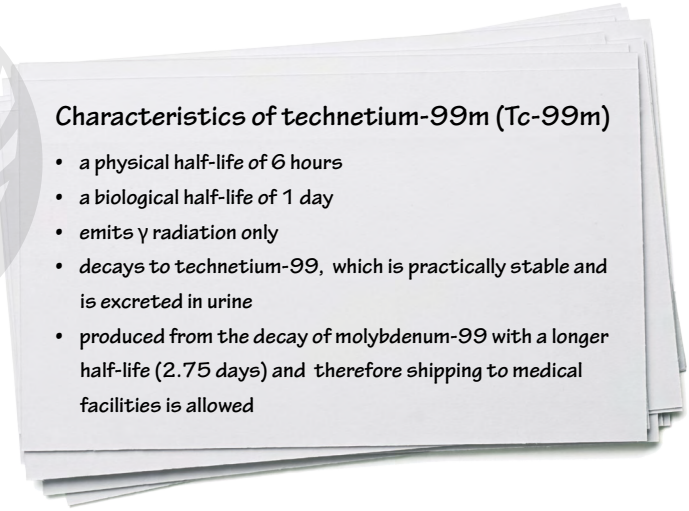


Fig 3.3g *Why Tc-99m is a suitable radionuclide for medical imaging.*

pharmacological 藥理性

DSE exam
12(2)Q4.7, 13(2)Q4.6, 19(2)Q4.6

3 Practical cases of radionuclide imaging

When doctors look at radionuclide images of patients, they look for ***cold spots*** and ***hot spots*** of the organs in the images. They indicate the parts of the organs which are possibly not functioning normally.

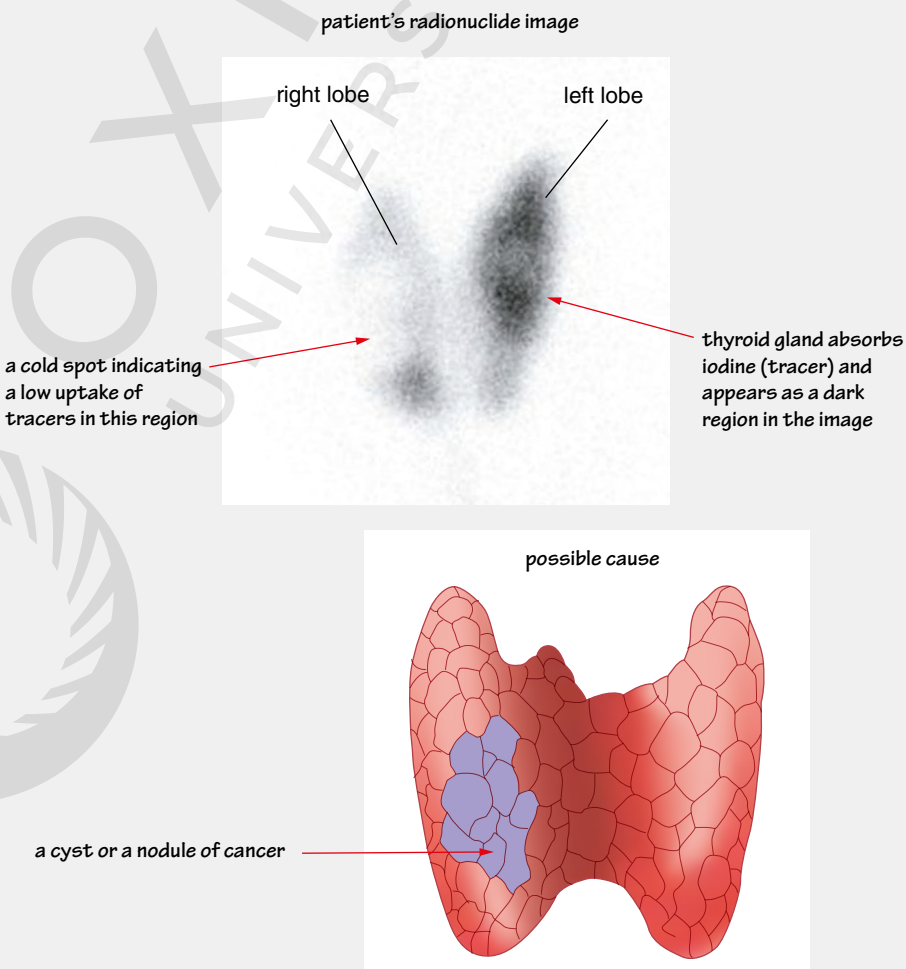
Cold spot: a region where the tracer is partially taken up in the organ and is shown brighter than normal in the image

Hot spot: a region where the tracer is taken up in excess and is shown darker than normal in the image

Let us look at the following cases.

Case 1

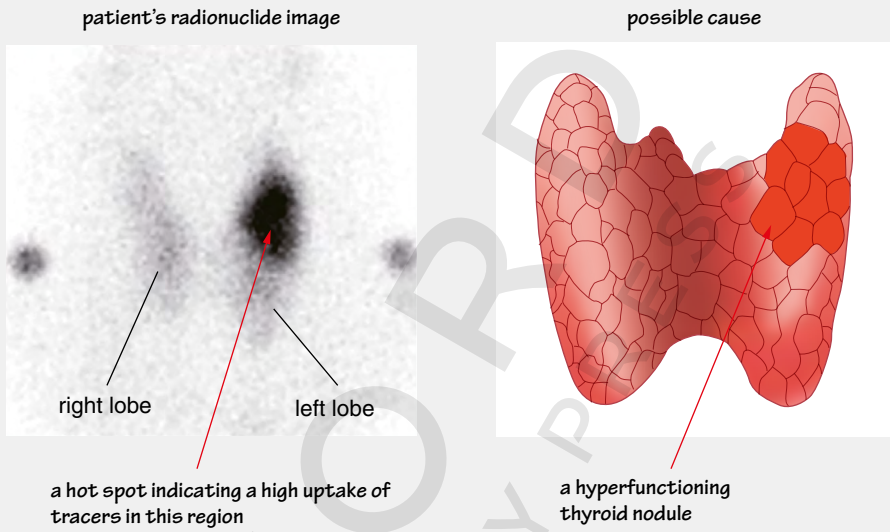
(Thyroid imaging with iodine as tracer)



cold spot 冷點　　hot spot 熱點

Case 2

(Thyroid imaging with iodine as tracer)



Google Forms Microsoft Forms

Quick check 9

→ E-version in OUP Exercise Platform is also available.

1 Which of the following about the radioactive sources used in radionuclide imaging is correct?

A Sources emitting only α radiation should be used.

B Sources emitting only γ radiation should be used.

C Sources emitting only β and γ radiation should be used.

D Sources emitting α, β and γ radiation can be used.

2 Which of the following about a cold spot of an organ in a radionuclide image is/are correct? Put a tick in the corresponding box(es).

☐ It is a white region in the image.

☐ It is a region which appears brighter than normal in the image.

☐ It indicates the part which is possibly not functioning normally.

☐ It indicates the part which is colder than surroundings.

☐ It indicates the part which has a relatively low uptake of the radionuclide.

3 In general, the radionuclide used as a tracer in radionuclide imaging is preferred to have a half-life

A as short as possible.

B as long as possible.

C of a few hours.

D of a few days.

4 Which of the following about a hot spot of an organ in a radionuclide image is/are correct? Put a tick in the corresponding box(es).

☐ It is a black region in the image.

☐ It is a region which appears darker than normal in the image.

☐ It indicates the part which is possibly not functioning normally.

☐ It indicates the part which is hotter than surroundings.

☐ It indicates the part which is healthy.

☐ It indicates the part which has a relatively high uptake of the radionuclide.

DSE goal

Compare radionuclide planar images with X-ray radiographic images.

DSE exam

15(2)Q4.7, 16(2)Q.4(a), 17(2)Q4.8, 18(2)Q4.6, 19(2)Q4.4, 22(2)Q.4(b)

4 Comparing radionuclide imaging and other imaging methods

a Characteristics of radionuclide imaging

Radionuclide imaging is different from many other forms of imaging in the following two ways.

1 The **source of radiation is placed inside the patient's body** and therefore the radiation detected comes from inside the patient (Fig 3.3h). Radionuclide imaging is an **invasive imaging method**.

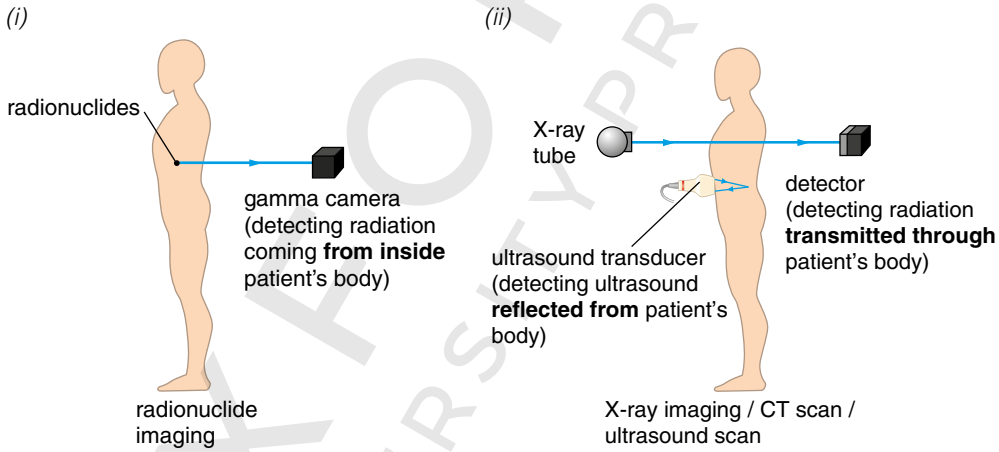


Fig 3.3h *(i) Internal source and (ii) external sources of radiation or ultrasound.*

2 It provides **functional information** but **very limited structural information** about the body parts (Fig 3.3i).

Providing functional information is a feature unique to radionuclide imaging and it is hence sometimes preferable to other imaging methods.

The functional information can help us identify and assess severity of diseases at an early stage. For example, a bone scan using radionuclide may identify small bone cancer earlier than other imaging methods.

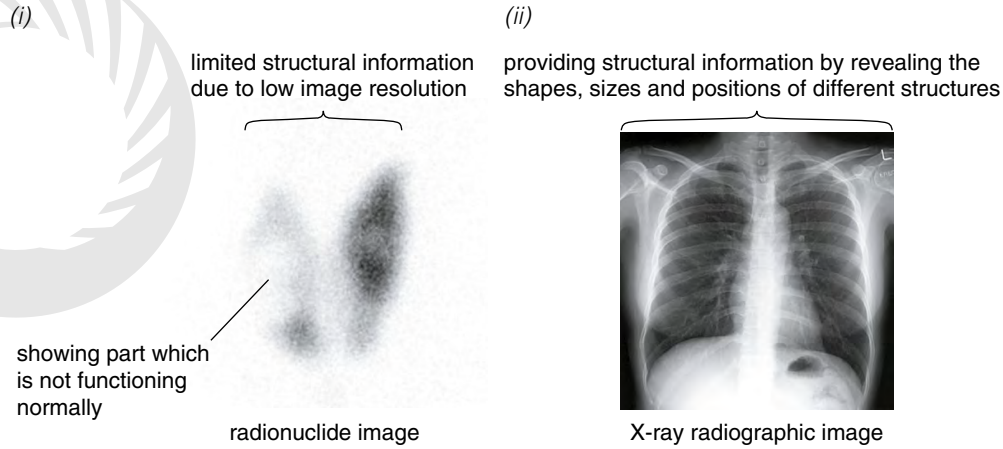


Fig 3.3i *Images (i) providing functional information but limited structural information, and (ii) providing structural information.*

b Comparing radionuclide imaging and X-ray radiographic imaging

Table 3.3a compares radionuclide imaging and X-ray radiographic imaging.

Aspect	Radionuclide imaging	X-ray radiographic imaging
Radiation source	☹ internal (invasive)	🙂 external (non-invasive)
Time taken for imaging	☹ longer	🙂 shorter
Exposure to ionizing radiation	☹ more	🙂 less
Resolution	☹ lower	🙂 higher
Information provided	🙂 mainly functional information ☹ limited structural information	🙂 mainly structural information ☹ no functional information
Major uses	to assess the severity of disease and identify disease at an early stage	to examine bone fractures and diagnose chest diseases

Table 3.3a *Comparison between radionuclide and X-ray radiographic imaging.*

Google Forms Microsoft Forms

Quick check 10

→ E-version in OUP Exercise Platform is also available.

1 When radionuclide imaging is compared with X-ray radiographic imaging, which of the following is/are correct? Put a tick in the corresponding box(es).

- ☐ Radionuclide images have a lower resolution than X-ray images.
- ☐ Radionuclide imaging is better for obtaining structural information about the target organ.
- ☐ Radionuclide imaging is better for obtaining functional information about the target organ.
- ☐ Radionuclide imaging takes a longer time for imaging.
- ☐ The patient exposes to less radiation in radionuclide imaging.

2 Which of the following imaging methods is/are non-invasive? Put a tick in the corresponding box(es).

- ☐ Ultrasound scanning
- ☐ X-ray radiographic imaging
- ☐ CT scanning
- ☐ Radionuclide imaging

Exercise 3.3

★**1** The following figure shows a kidney scan using technetium-99m as radionuclide tracer (Fig a).

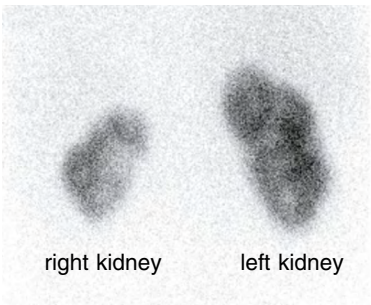


Fig a

Which of the following information can be deduced from the image?

A The right kidney consumes technetium more quickly than the left kidney.

B The right kidney absorbs less technetium than the left kidney.

C Technetium-99m decays faster in the right kidney than in the left kidney.

D Technetium-99m in the left kidney emits more radiation than that in the right kidney.

★**2** In radionuclide imaging, it is preferred for the radionuclide to have a half-life of a few hours so that

(1) the activity is not too low when they reach the target organ.

(2) the activity is low after a few days.

(3) they leave the body before the activity is too low.

A (2) only B (1) and (2) only

C (1) and (3) only D (2) and (3) only

★**3** Which of the following statements about radionuclide imaging is correct?

A It allows us to examine the structure of an organ in detail.

B It allows us to view an organ from different angles.

C It may help us identify a body tissue that is more active than normal.

D It may help us find a substance that is normally not present in the body.

4 (a) State the reason why a γ source should be used as tracers in radionuclide imaging.

(b) State a reason why the tracers in radionuclide imaging should not emit α or β radiation.

(c) Apart from the characteristic stated in (a) and (b), state two other characteristics that the tracers should have.

5 Compare radionuclide imaging and X-ray radiographic imaging in terms of the time taken for imaging. Explain your answer briefly.

6 Compare radionuclide imaging and X-ray radiographic imaging in terms of the exposure to ionizing radiation. Explain your answer briefly.

★**7** The radioisotope of cobalt, Co-60, has a physical half-life of 5.27 years and a biological half-life of 10 days. (Take 1 year = 365.25 days.)

(a) What is the difference between physical half-life and biological half-life?

(b) Find the effective half-life of Co-60.

(c) If a certain amount of Co-60 is taken into the body, how long does it take for the amount of Co-60 left in the body to reduce to 25% of its original intake?

★**8** Radioactive technetium-99m has a physical half-life of 6 hours and a biological half-life of 24 hours. It is commonly used as tracer for radionuclide imaging of the brain.

(a) Find the effective half-life of technetium-99m.

(b) During radionuclide imaging, technetium-99m with an initial activity of 12 Bq is injected into a patient's body. Find the activity of the radionuclide inside the body 2 days after the injection.

★**9** Figure b shows a radionuclide image of the bones of a patient.



Fig b

Locate a region in the image at which a bone disease may occur. Explain your answer.

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3.4 Health risks of imaging methods involving ionizing radiation

In this chapter, we have discussed X-ray radiographic imaging, CT scanning and radionuclide imaging. These imaging methods involve the use of ionizing radiation, which can damage living cells. When choosing an imaging method, it is important to take this risk into consideration and take appropriate precautions when using it.

DSE goal

Discuss the health risk and safety precautions for ionizing radiation.

1 Biological effects

Recall from Book 5 Chapter 1 that ionizing radiation can damage living cells, and may lead to various effects such as skin damage, cataract and cancer.

It is worth noting that pregnant women should not use imaging methods that involve ionizing radiation (Fig 3.4a) unless there are very special reasons. There is a lot of cellular activity in the foetus as the body is being formed. Therefore, an unborn child is at even higher risks from their harmful effects than other people.



Fig 3.4a *Pregnant women should use imaging methods that do not involve ionizing radiation.*

DSE goal

Compare effective dose in diagnostic medical procedures involving ionizing radiation.

DSE exam

17(2)Q.4, 20(2)Q.4(c)

2 Effective dose

We can estimate the risks to health of these imaging methods by calculating the **effective dose**, in the unit of sievert (Sv), for the whole body. Its value is calculated according to the amount of ionizing radiation received, the types of ionizing radiation received and the chances for ionizing radiation to induce cancers in different organs and tissues.

effective dose 有效劑量

Know more

Equivalent dose and effective dose

In Book 5 Chapter 1, we have learned the equivalent dose.

The calculation of equivalent dose takes the biological effectiveness of different types of radiation into account (Fig a). The equivalent dose is equal to the sum of the products of the radiation weighting factor and the absorbed dose of each type of radiation.

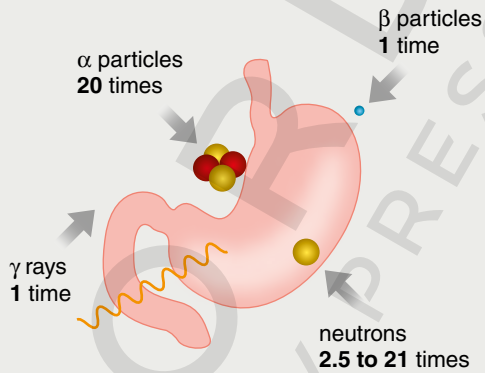


Fig a *Radiation weighting factors of different types of radiation.*

The calculation of effective dose takes into account the type of radiation and also the nature of each tissue or organ being irradiated (Fig b). The effective dose is equal to the sum of the products of the tissue weighting factor and the equivalent dose of each tissue or organ.

The following shows an example of calculating the effective dose.

When all parts of the body receive ionizing radiation so that each part has an equivalent dose of 1 mSv,

Effective dose =

0.04×1 mSv (for liver)

$+ 0.12 \times 1$ mSv (for colon)

$+ 0.12 \times 1$ mSv (for lungs)

$+ 0.12 \times 1$ mSv (for stomach)

$+ 0.01 \times 1$ mSv (for skin)

+ ...

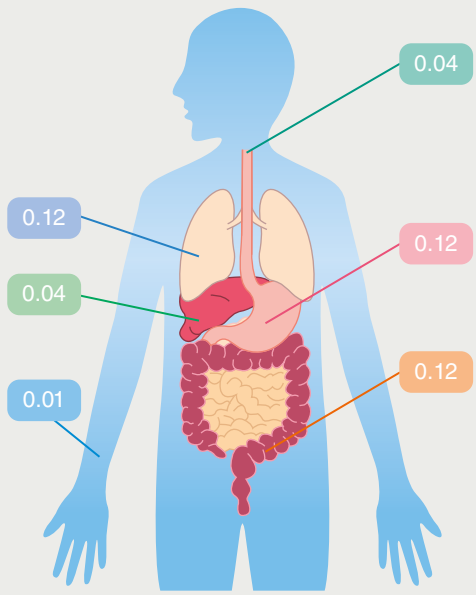


Fig b *Tissue weighting factors of different tissues or organs.*

Table 3.4a shows the effective doses received from different imaging procedures.

Imaging method	Diagnostic procedure	Effective dose / mSv
X-ray radiographic imaging	Chest	0.02
	Abdomen	0.7
	Spine	1.5
	Barium study	3
CT scanning	Head	2
	Spine	6
	Chest	8
	Abdomen / pelvis	10
Radionuclide imaging	Chest (lung ventilation)	0.1
	Kidney	1
	Thyroid	1
	Bone	4

Table 3.4a *Effective doses of different medical imaging procedures.*

From Table 3.4a, we can see that the effective dose from CT scanning is generally higher than that from X-ray radiographic imaging and radionuclide imaging.

3 Radiation protection

In order to minimize the risk of radiation exposure in medical imaging involving ionizing radiation, appropriate protection measures should be taken. Some radiation protection measures are listed as follows:

1 **Time**: Avoid unnecessary radiation examinations. In addition, the duration of imaging procedures should not be longer than necessary.

✎ Ts may remind Ss of the inverse square law.

2 **Distance**: Keep a distance from the radioactive source unless undergoing an imaging procedure.

3 **Shielding**: Barriers should be used to shield radiation. All radioactive sources should be handled in lead shield or under water, or by remote control in rooms constructed of thick concrete or lined with lead. Wear protective clothing during imaging procedures.

4 **Containment**: Make sure the radioactive sources are kept out of the environment. Rooms should have an air pressure lower than outside so that contaminated air does not leak out of the room.

Exercise 3.4

1 Under normal situation, which of the following imaging methods is most suitable for inspecting a foetus?

A Ultrasound scanning
B X-ray radiographic imaging
C CT scanning
D Radionuclide imaging

2 In general, which of the following imaging methods has the highest effective dose absorbed by the patient?

A Ultrasound scanning
B X-ray radiographic imaging
C CT scanning
D Radionuclide imaging

*3 Different weighting factors are given to different types of ionizing radiation in the calculation of effective dose. α radiation is given a much larger weighting factors than the others because it has the

A lowest penetrating power.
B highest penetrating power.
C weakest ionizing power.
D strongest ionizing power.

*4 Which of the following is/are valid method(s) for reducing radiation exposure? Put a tick in the corresponding box(es).

☐ Avoid taking unnecessary X-ray scans
☐ Drink plenty of water
☐ Wear lead apron
☐ Surround X-ray rooms with thick concrete walls
☐ Keep radioactive sources in low-pressure rooms

*5 When taking an X-ray radiographic image, the imaging process is conducted in a room with thick concrete walls (Fig a). The patient has to wear a protective clothing. The operator controls the instruments in a separate room. Explain how these measures reduce the risk of radiation exposure.

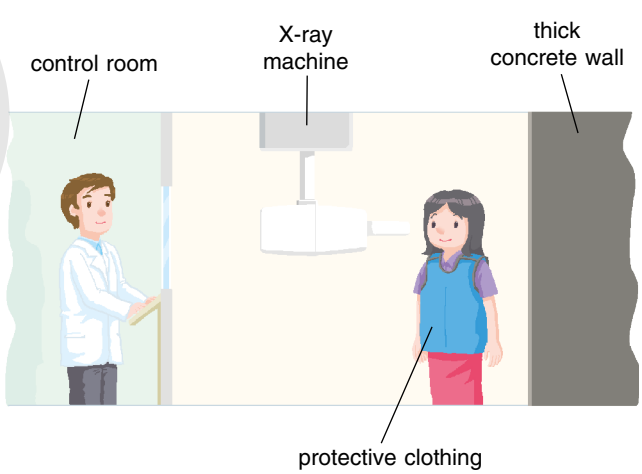
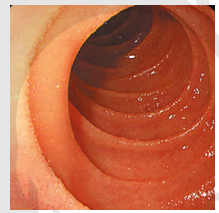
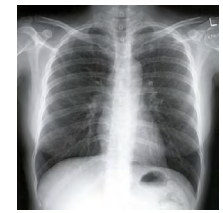
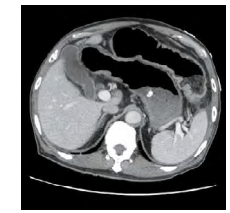
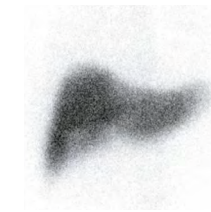


Fig a

Comparison of different imaging methods

In this book, we have gone through different types of medical imaging methods to help us look into different parts and different properties of our bodies. Let us summarize what we have learned in the following table.

	Fibre optic endoscopy	Ultrasound scanning	X-ray radiographic imaging	CT scanning	Radionuclide imaging
					
Radiation source	• Visible light • External • Non-ionizing	• Ultrasound • External • Non-ionizing	• X-ray • External • Ionizing	• X-ray • External • Ionizing	• γ radiation • Internal • Ionizing
Invasive	Yes	No	No	No	Yes
Image	• Produced from visible light reflected from body tissues • Shows interior surface of organ / tissue • Medium resolution	• Produced from intensity of ultrasound reflected at tissue boundaries • Shows cross-sectional view • High resolution • Dark area implies acoustic shadow or small difference in acoustic impedance	• Produced from intensity of X-rays passing through body • Shows planar view • High resolution • Darker area implies smaller total attenuation (higher X-ray intensity)	• Produced by back projection of attenuation profiles • Shows cross-sectional view • Medium resolution • Darker area implies smaller linear attenuation coefficient	• Produced from γ radiation emerging from body • Shows planar view • Low resolution • Darker area implies more radionuclide contained in area / organ

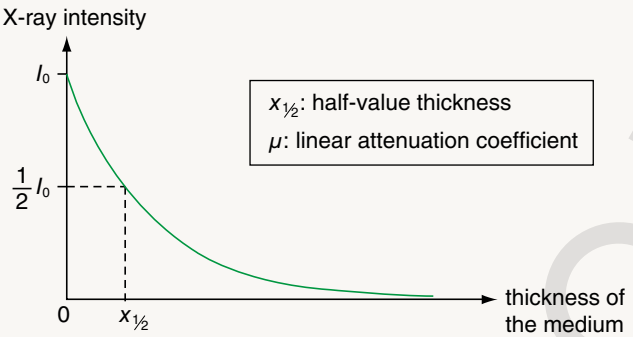
Application	• Examine internal surface of hollow organs • Sample tissues or perform simple operations • Minimally invasive surgery	• Examine body tissues, determine location and size of structure or disease • Commonly used for monitoring foetus during pregnancy	• Examine body tissues, determine location and size of structure or disease • Commonly used to quickly diagnose broken bones and lung diseases	• Examine body tissues, determine location and size of structure or disease • Commonly used for investigating bones and chest diseases	• Determine location and shape of organs • Assess function of organs • Assess severity of disease and identify disease at early stages
Pros	• Readily accessible • Real-time imaging • Allow tissue sampling and simple operations	• Readily accessible • Real-time imaging • No discomfort	• Widely available • Cheap • Fast	Possible to perform multi-planar reformation and 3-D reconstruction	Can provide functional information of organs
Cons	• Limited field of view • Require anaesthesia / fasting • May lead to perforation / complications	• Limited field of view • Limited tissue penetration	• Overlapping of structures may occur • Limited contrast between soft tissues • Potential allergic reaction to artificial contrast medium • Exposure to ionizing radiation	• High cost • High radiation dose	• High cost • Delayed image reproduction • Low resolution • Exposure to ionizing radiation

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Key concepts 3

1 Attenuation of X-rays

X-rays lose energy when travelling through a medium.



Related formulae

$$I = I_0 e^{-\mu x}$$

$$I = I_0\left(\frac{1}{2}\right)^n$$

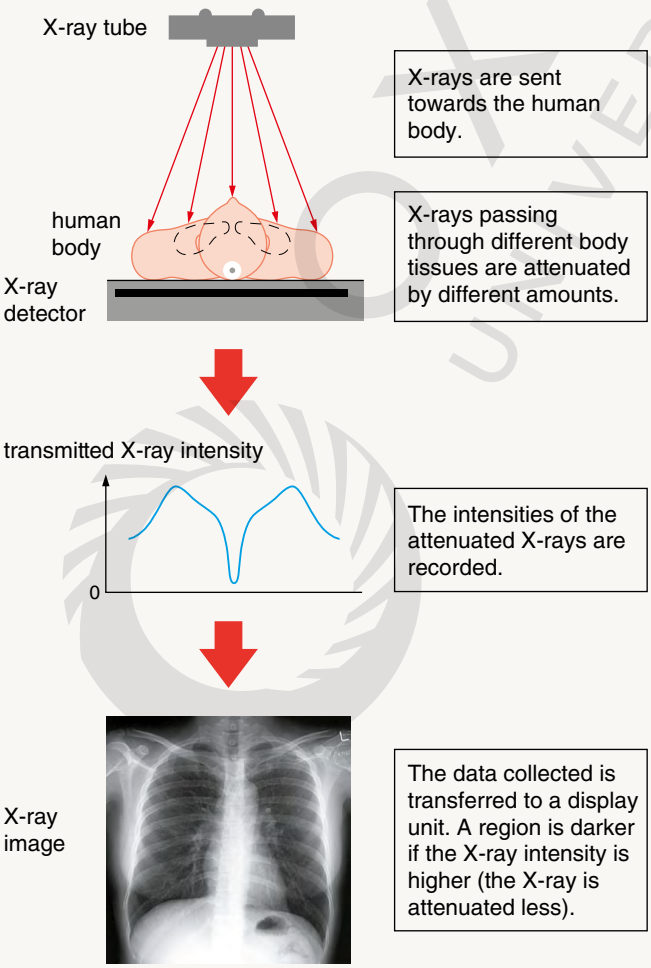
$$x_{½} = \frac{\ln 2}{\mu}$$

Factors affecting attenuation

frequency of X-ray ↑ $\Rightarrow \mu \downarrow$

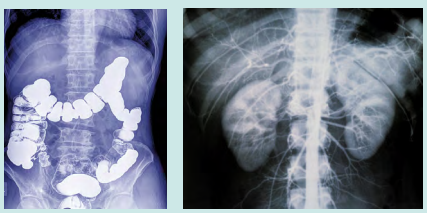
density of medium ↑ $\Rightarrow \mu \uparrow$

2 X-ray radiographic imaging



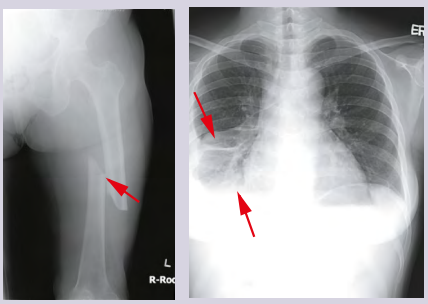
Artificial contrast media

- high linear attenuation coefficient
- makes an organ appear lighter than normal in an X-ray image



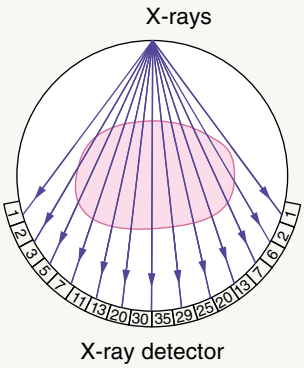
Practical uses

- inspect bone fractures
- diagnose lung diseases

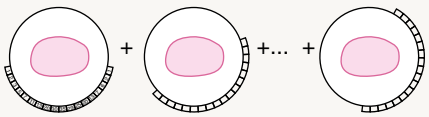


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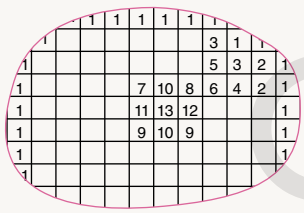
3 CT scanning



X-rays are sent towards the human body. The recorded intensities are processed to obtain an attenuation profile.

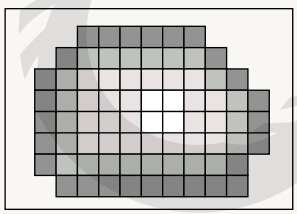


As the X-ray tube and the detector rotate, a number of attenuation profiles are obtained.



map of attenuation coefficients

The computer uses back projection algorithm to reconstruct a map of attenuation coefficients of the cross-section of the body.

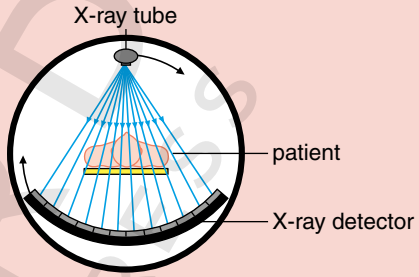


CT image

The map is filled different shades of grey to form a CT image.

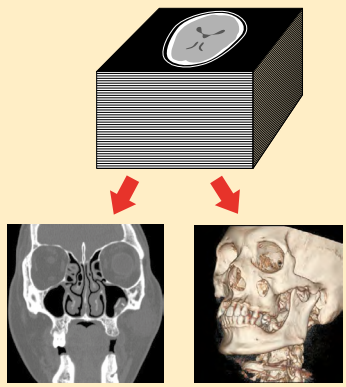
CT scanner

- X-ray tube and X-ray detector rotated in a circle
- patient is slowly moved through the scanner



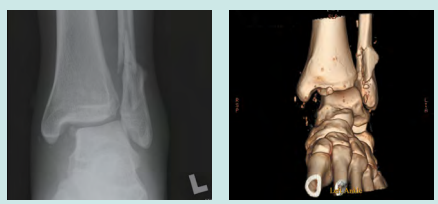
Digital post-processing

A CT scanner produces a stack of tomographs which can be used to perform multi-planar reformation or 3-D reconstruction.



Compare with X-ray radiography

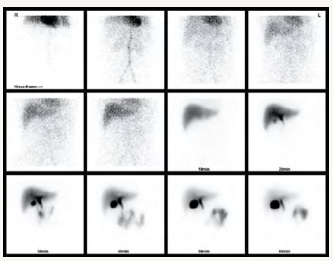
- Radiography: 2-D map showing total attenuation
- CT scanning: 3-D map of linear attenuation coefficients



4 Radionuclide imaging



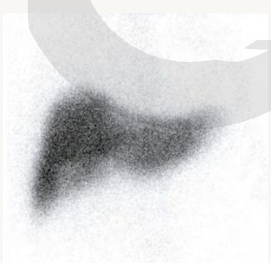
A small amount of radionuclide tracers are injected into the patient's body.



The tracers circulate inside the body and accumulate in the target organ after some time.



A gamma camera detects the radiation emitted from the radionuclides.



The resulting image shows the distribution of radionuclides inside the body.

Half-lives

Physical half-life (t_p)

= time taken for half of the nuclei to undergo radioactive decay

Biological half-life (t_b)

= time taken for half of the tracers to leave the body by biological processes

Effective half-life (t_e)

= time taken for the activity of tracers in the body to decrease by half

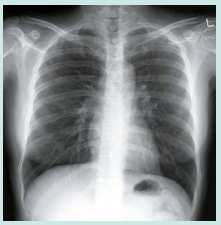
$$\frac{1}{t_e} = \frac{1}{t_p} + \frac{1}{t_b}$$

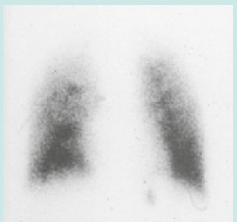
Choosing radionuclides

1. emit γ radiation only
2. γ radiation emitted has suitable energy
3. suitable half-life
4. non-toxic and no pharmacological effect
5. can be combined with a wide range of substances
6. decay to a stable nuclide
7. readily available and inexpensive

Compare with X-ray radiography

- Radiography: provides detailed structural information
- Radionuclide imaging: provides functional information (organ may not function normally if an area is brighter or darker than normal)





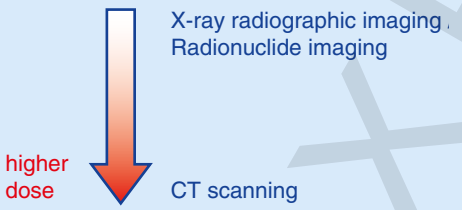
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5 Health risks of imaging methods involving ionizing radiation

- Ionizing radiation can damage living cells.
- Pregnant women should not use imaging methods involving ionizing radiation in general.

Effective dose

- Equivalent dose corrected for the effect of radiation on different organs
- Measured in sievert (Sv)
- In general,



Radiation protection

Time:

Avoid unnecessary radiation examinations

Distance:

Keep a distance from the radioactive source

Shielding:

Use barriers to shield radiation

Containment:

Keep radioactive sources out of the environment

Key terms

1 artificial contrast medium 人工顯影劑 p.129
2 attenuation profile 衰減輪廓 p.137
3 back projection algorithm 反投影算法 p.138
4 biological half-life 生物半衰期 p.152
5 computed tomographic image 電腦斷層影像 p.136
6 computed tomography 電腦斷層造影 p.136
7 effective dose 有效劑量 p.160
8 effective half-life 有效半衰期 p.152
9 half-value thickness 半值厚度 p.118
10 linear attenuation coefficient 線衰減係數 p.120
11 physical half-life 物理半衰期 p.152
12 radionuclide imaging 放射性核素造影術 p.148
13 tracer 示蹤物 p.149
14 X-ray radiographic image X 射線放射攝影圖像 p.116

Revision 3

Multiple-choice questions

★1 3.1 X-ray is useful in medical imaging because of its

A electromagnetic nature.

B high energy.

C ionizing property.

D penetrating property.

★2 3.1 The graph shows the intensities of an X-ray beam after travelling different distances through a medium (Fig a). Estimate the intensity of the X-ray after travelling for 8 cm.

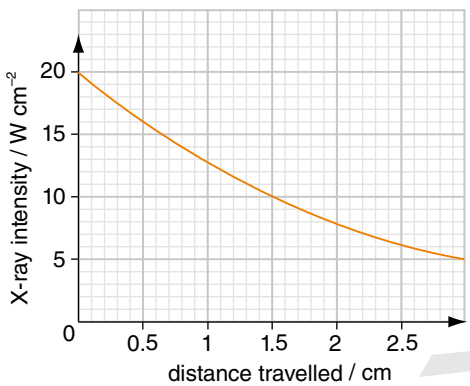


Fig a

A 0.0781 W cm^{-2}

B 0.496 W cm^{-2}

C 0.521 W cm^{-2}

D 3.75 W cm^{-2}

★3 3.1 Which of the following statements about X-ray radiographic imaging is/are correct?

(1) It produces images by measuring the intensity of X-rays passing through a patient's body.

(2) Barium is used as an artificial contrast medium because it emits ionizing radiation.

(3) The body structures that reflect most X-rays appear white in an X-ray image.

A (1) only

B (1) and (2) only

C (2) and (3) only

D (1), (2) and (3)

★4 3.1 Figure b shows the change in intensity I when an X-ray beam travels in two gases P and Q.

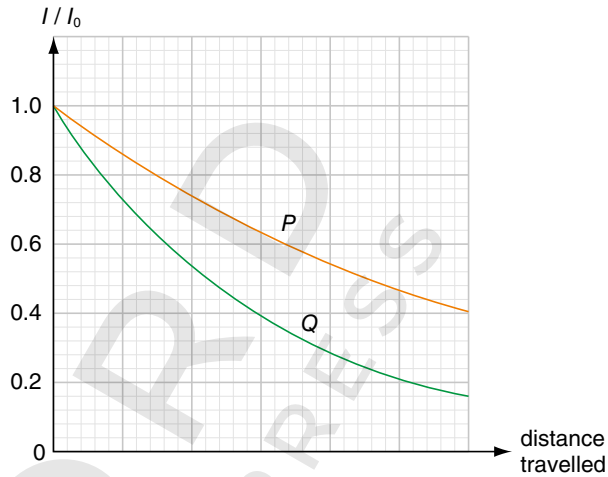


Fig b

P and Q are gases of the same element but with different densities. Which of the following statements is/are correct?

(1) The half-value thickness of P is smaller than that of Q.

(2) The linear attenuation coefficient of P is smaller than that of Q.

(3) The density of P is lower than that of Q.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

★5 3.2 Which of the following is/are the advantage(s) of CT scanning over X-ray radiographic imaging?

(1) The patient receives less ionizing radiation.

(2) The function of an organ can be assessed.

(3) It can provide three-dimensional images of body tissues.

A (3) only B (1) and (2) only

C (1) and (3) only D (1), (2) and (3)

★6 Misc A patient is suspected to have liver failure (i.e. the liver cannot function properly). Which of the following imaging methods is the most suitable for making a diagnosis?

A Ultrasound scanning

B Radionuclide imaging

C Fibre optic endoscopy

D CT scanning

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*7 Misc The figures below show two images of a gall bladder using different imaging techniques (Fig c and d).

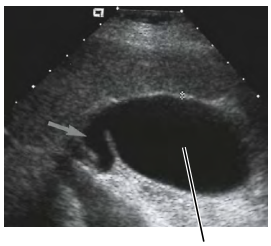


Fig c

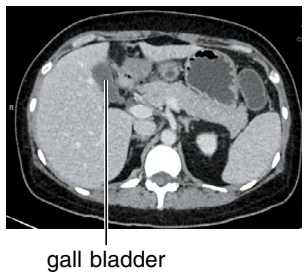


Fig d

Which of the following statements about the two images is/are correct?

(1) Both images are taken using ionizing radiation.

(2) The dark areas in both images indicate that the corresponding regions absorb more radiation.

(3) Both images show the cross-sections of the abdomen.

A (3) only

B (1) and (2) only

C (2) and (3) only

D (1), (2) and (3)

*8 3.3 The biological half-life of a radionuclide is much longer than its physical half-life. Which of the following statements is/are correct?

(1) The effective half-life of the radionuclide is approximately equal to the physical half-life.

(2) Before most of the radionuclides has decayed, only a small proportion has been removed from the body.

(3) This radioisotope is suitable for medical imaging even if it decays to an unstable nuclide.

A (1) only

B (1) and (2) only

C (2) and (3) only

D (1), (2) and (3)

*9 Misc Which of the following statements about CT scanning is/are correct?

(1) A CT image is produced by using the back projection algorithm on the attenuation profiles.

(2) A CT image is a map of the distribution of sources of X-rays in the body.

(3) It is generally used to monitor unborn babies during pregnancy.

A (1) only

B (1) and (2) only

C (2) and (3) only

D (1), (2) and (3)

*10 Misc The following figure shows a parallel beam of X-ray passing through an arm.

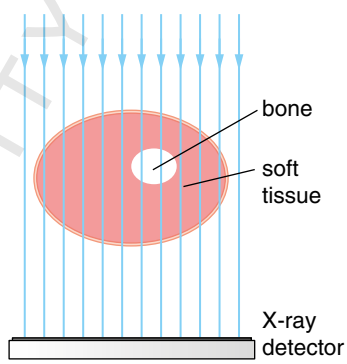
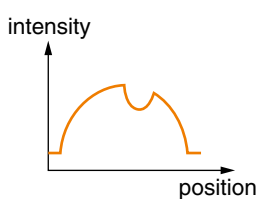


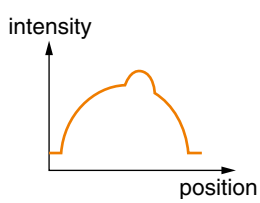
Fig e

Which of the following graph shows the most possible intensity of X-ray detected at different positions across the detector?

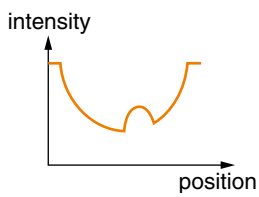
A



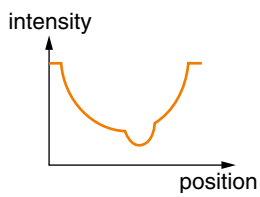
B



C



D



★**11** Misc Which of the following correctly compare(s) radionuclide tracer in radionuclide imaging and artificial contrast medium in X-ray radiographic imaging?

	Radionuclide tracer	Artificial contrast medium
(1)	accumulates in specific organ	spreads evenly inside body
(2)	appears black on image	appears white on image
(3)	emits ionizing radiation	absorbs ionizing radiation

A (1) and (2) only B (1) and (3) only
C (2) and (3) only D (1), (2) and (3)

12 3.3 **HKDSE 2013 Paper 2 Q4.6**

Technetium-99m is a radioisotope which undergoes γ decay with a half-life of 6 hours. Some technetium-99m was combined with a substance which can be easily absorbed by liver. The compound was taken by a patient and a series of images were taken by a gamma camera at different times (Fig f). Which of the following statements is/are correct?

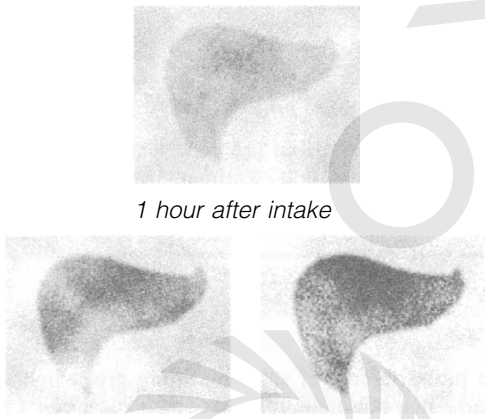


Fig f

(1) The darker part of the images corresponds to the part of the liver causing a greater attenuation of γ rays.
(2) This series of images provides functional information about the liver of the patient.
(3) The difference between the images is solely due to the decay of technetium-99m.

A (1) only B (2) only (36%)
C (1) and (3) only D (2) and (3) only

13 3.1 **HKDSE 2014 Paper 2 Q4.6**

An object is made up of two different materials P and Q of 1 cm equal thickness as shown in Figure g. The linear attenuation coefficients of P and Q for X-rays are 0.05 cm^{-1} and 0.68 cm^{-1} respectively. An X-ray beam of intensity I_0 is incident on the object and emerges from the object with an intensity I. Which of the following expressions gives the ratio $\frac{I}{I_0}$?

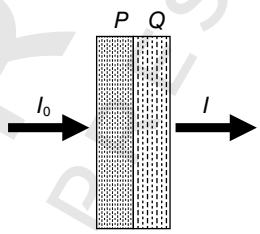


Fig g

A $\frac{0.05}{0.68}$

B $\frac{(0.68 - 0.05)^2}{(0.68 + 0.05)^2}$

C $e^{-\frac{0.05}{0.68}}$

D $e^{-(0.05 + 0.68)}$ (44%)

14 3.1 **HKDSE 2016 Paper 2 Q4.7**

An X-ray beam of intensity I_0 is incident on a medium of linear attenuation coefficient μ. After travelling a distance x in that medium as shown, the intensity of the beam becomes I. A graph of $\ln\left(\frac{I}{I_0}\right)$ is plotted against x. What does the magnitude of the slope of the graph represent?

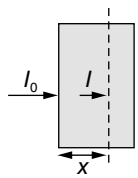


Fig h

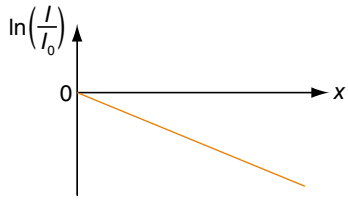


Fig i

A $\frac{\mu}{2}$ B $\frac{\ln 2}{\mu}$

C $\frac{1}{\mu}$ D μ (49%)

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Conventional questions

★**15** 3.1 An X-ray beam of initial intensity 8×10^{-2} W m^{-2} is travelling in an aluminium block with a linear attenuation coefficient of 0.99 cm^{-1}.

(a) Find the thickness of the aluminium block required to reduce the intensity of the X-ray beam to 3.8×10^{-2} W m^{-2}. (2 marks)

(b) (i) State the meaning of half-value thickness of a material for X-rays. (1 mark)

(ii) Find the half-value thickness of the aluminium block for X-rays. (2 marks)

(iii) Lead has a larger linear attenuation coefficient than aluminium. Compare the half-value thickness of lead and aluminium. (1 mark)

★**16** 3.3 (a) State the meanings of physical half-life and biological half-life. (2 marks)

(b) Iodine-125 (I-125) is a tracer suitable for radionuclide imaging. It has a physical half-life of 60 days and an effective half-life of 16 days.

(i) Find the biological half-life of I-125. (2 marks)

(ii) Two identical I-125 samples P and L have the same initial activity A_0. Sample P is introduced into a patient's body while sample L is kept in a laboratory. In Figure j, sketch two curves to show how the activities of P in the patient's body and L in the laboratory change with time. (2 marks)

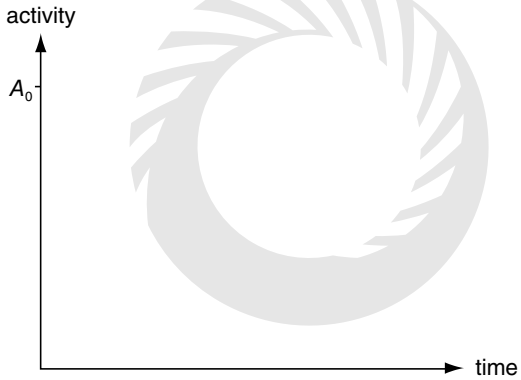


Fig j

(iii) Explain what type(s) of nuclear radiation I-125 should emit. (2 marks)

★**17** 3.1 Figure k shows an X-ray radiographic image of the chest. The circled light grey area shows that there is an irregular mass in the lungs.

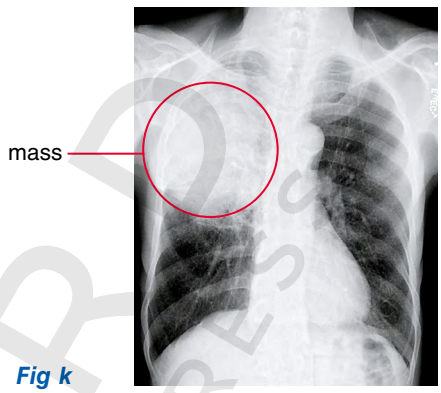


Fig k

(a) (i) How is the shades of grey of an area on an X-ray radiographic image related to the attenuation of the X-rays? (1 mark)

(ii) Explain why the light grey area in the image suggests that there is a mass. (2 marks)

(b) Explain why X-ray radiographic imaging is a good tool for first-line screening when there is an outbreak of infectious lung disease. (2 marks)

★**18** 3.2 Figure l shows a CT image of the brain.

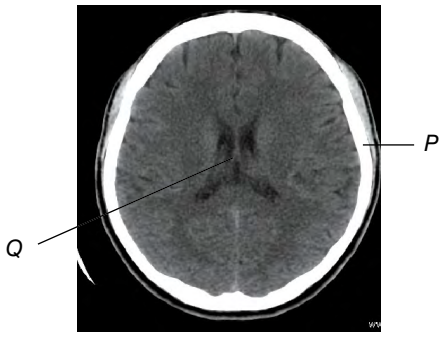


Fig l

(a) Describe how a CT image of a slice of the body is produced. (4 marks)

(b) From the image of the brain, what can you say about the difference between the tissues at P and Q? (1 mark)

(c) The image consists of 512×512 pixels with each pixel size 0.391 mm^2. What are the dimensions of the image? (2 marks)

(d) State one reason why modern CT images have a higher resolution than early ones. (1 mark)

★**19** Misc Amanda has a severe pain in her forearm after a car accident. A doctor sends her to take a radiographic image.

(a) Explain why the doctor askes Amanda to remove her metal bracelet before taking the image. (2 marks)

(b) Figure m shows the radiographic image of Amanda's forearm.



Fig m

Describe briefly how the image is formed. (2 marks)

(c) Artificial contrast medium is sometimes used in radiographic imaging.

(i) State one property of an artificial contrast medium. (1 mark)

(ii) Is it necessary to use an artificial contrast medium in Amanda's case? Explain your answer. (2 marks)

(d) State one health risk of receiving high doses of X-rays. (1 mark)

★**20** Misc Figure n shows a CT image of the heart using iodine as an artificial contrast medium. The white dots indicated by arrows shows blood vessels on the surface of the heart.

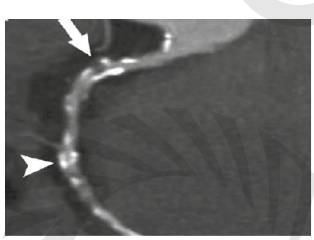


Fig n

(a) Iodine is injected into the bloodstream before the scan. Explain how it can make the blood vessels highlighted in the CT image. (2 marks)

(b) The doctor can find out if the patient suffers from coronary artery diseases by measuring the diameter of the blood vessels. Explain why this cannot be done with X-ray radiographic imaging. (1 mark)

(c) State a disadvantage of using CT scanning to diagnose heart diseases. (1 mark)

★**21** Misc Jimmy has a brain examination. A small amount of radionuclide technetium-99m (Tc-99m) is introduced into Jimmy's body for taking image of his brain.

(a) Describe briefly how a radionuclide image is produced. (3 marks)

(b) Suggest a reason why Jimmy does not take images immediately after the injection of Tc-99m. (1 mark)

(c) Tc-99m has a physical half-life of 6 hours and a biological half-life of 1 day. Find the effective half-life of Tc-99m. (2 marks)

(d) Jimmy also has an X-ray radiographic imaging for his head. Figure o shows the image obtained.

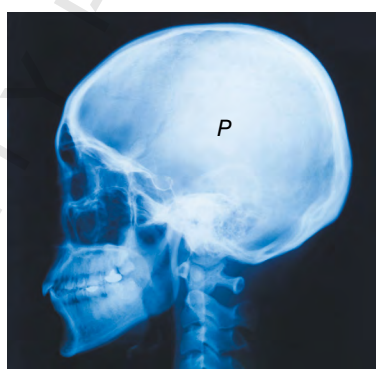


Fig o

(i) Explain briefly why region *P* appears white in colour. (2 marks)

(ii) Compare radionuclide imaging and X-ray radiographic imaging in terms of the information provided by the images. (2 marks)

★**22** Misc Connie takes two medical images of her liver. Figure p shows the image from CT examination and Figure q shows the one from ultrasound scanning.



Fig p

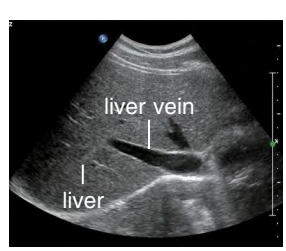


Fig q

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(a) (i) Explain how a tomograph is produced in CT scanning. (3 marks)

(ii) Explain how the image in Figure p is constructed from tomographs. (2 marks)

(b) State what the light areas indicate in each image. (2 marks)

(c) State one advantage of ultrasound scanning over CT scanning. (1 mark)

★★ **23** 3.1 An X-ray beam passes through a medium with linear attenuation coefficient μ. Its initial intensity is I_0. After travelling a distance x in the medium, its intensity is given by

$$I = I_0 e^{-\mu x}$$

(a) The half-value thickness $x_{½}$ is another useful quantity to assess the ability of a medium to attenuate X-rays.

(i) State the meaning of half-value thickness of a material for X-rays. (1 mark)

(ii) Show that $x_{½} = \dfrac{\ln 2}{\mu}$. (2 marks)

(b) Figure r shows the variation of $\ln\left(\dfrac{I}{I_0}\right)$ against x.

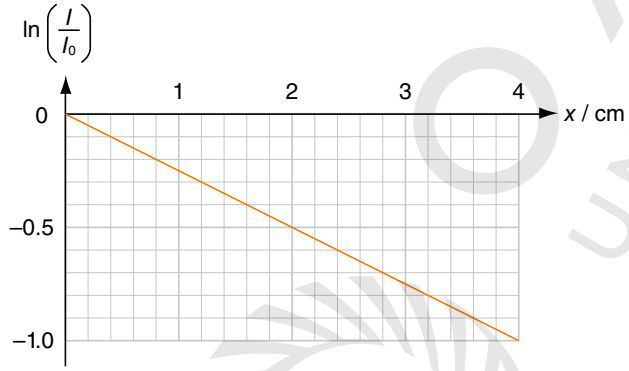


Fig r

(i) Derive an expression for the slope of the graph. Hence, find μ for this medium. (3 marks)

(ii) Sketch a new graph in Figure r if the density of the medium becomes higher. (1 mark)

Refer p.118

See p.178 for the Exam Report.

24 3.3 **HKDSE 2016 Paper 2 Q.4**

(a) Images *A*, *B* and *C* in Figure s were obtained by different medical imaging methods.

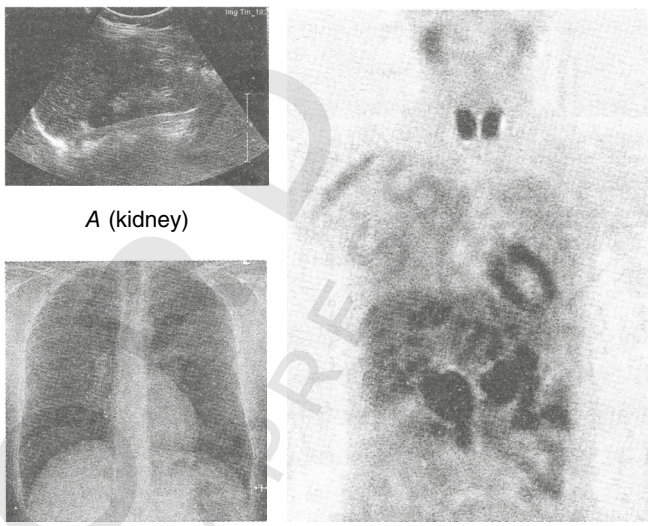


Fig s

(i) Which one is produced by radionuclide imaging? Explain how this image is formed. No need to describe the structure and mechanism of the detecting instrument used. (4 marks)

(ii) State **one** advantage of radionuclide imaging over the other two imaging methods. (1 mark)

(b) A radioactive isotope of initial amount N_0 decays to become N_t after time t. The graph below shows the variation of the ratio $\dfrac{N_t}{N_0}$ with time t.

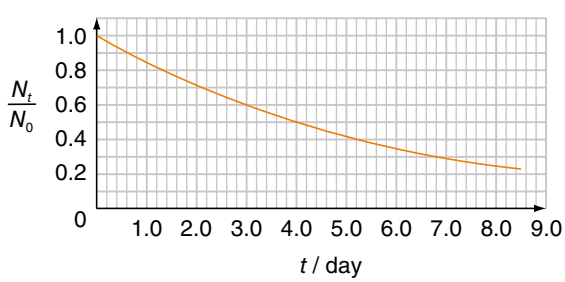


Fig t

(i) Use the graph to find the half-life of the radioactive isotope. (1 mark)

A chemical compound containing this radioactive isotope is used as a 'tracer' for injecting into a patient to study a physiological process. The biological half-life of this 'tracer' is 2 days.

(ii) What is meant by the **biological half-life** of the 'tracer'? (1 mark)

(iii) If 50 mg of this 'tracer' is injected initially, estimate the time taken for the amount of this **radioactive** chemical compound **remaining** in the body to drop to 10 mg. (3 marks)

See p.178 for the Exam Report.

25 HKDSE 2017 Paper 2 Q.4

Misc

X-ray radiographic imaging and computed tomography (CT) scans are used for medical purposes.

(a) Briefly describe how X-ray is produced. (1 mark)

(b) State an advantage of a CT scan over X-ray radiographic imaging. (1 mark)

(c) The effective dose of radiation absorbed can be measured in millisieverts (mSv) or expressed as the time taken to receive the equivalent dose from background radiation. The effective doses for a chest X-ray radiographic imaging and a chest CT scan are shown below.

	Effective dose (mSv)	Equivalent background radiation dose (days)
Chest X-ray radiographic imaging	0.02	1.85
Chest CT scan	6.6	610.5

Table a

(i) Briefly explain why the effective dose of a CT scan is much higher. (1 mark)

(ii) A head CT scan has an effective dose of 1.5 mSv. Based on the information from the table, estimate its equivalent background radiation dose. (1 mark)

(d) In a CT scan, a narrow X-ray beam of initial intensity I_0 transmits through lung cavity, soft tissue and bone along its path. The table below shows the linear attenuation coefficients of the tissues, and the path lengths of the X-ray in the tissues.

	Linear attenuation coefficient (cm^{-1})	Path length (cm)
Lung cavity	0.1	19.8
Soft tissue	0.18	8.8
Bone	0.48	4.4

Table b

(i) Briefly explain the large difference in linear attenuation coefficient between lung cavity and bone. (1 mark)

(ii) Determine the value of $\frac{\text{transmitted intensity } I}{\text{initial intensity } I_0}$ of the X-ray after transmitted through lung cavity, soft tissue and bone. (3 marks)

(e) A student suggests that a CT scan can be used for checking a foetus. Briefly explain whether you agree or not. If you do not agree, suggest a suitable medical imaging method for checking a foetus. (2 marks)

26 HKDSE 2020 Paper 2 Q.4

Misc

(a) State briefly how X-rays are produced. (1 mark)

(b) The table below shows the linear attenuation coefficients of soft tissue and bone for an X-ray beam.

	Linear attenuation coefficient
Soft tissue	$\mu_s = 0.51\ cm^{-1}$
Bone	$\mu_b = 2.46\ cm^{-1}$

Table c

(i) State one factor contributing to a higher linear attenuation coefficient of bone compared to soft tissue. (1 mark)

(ii) X-ray beam of intensity I_0 passes through soft tissue that is 5.6 cm in thickness and is attenuated to intensity I. What thickness of bone would yield the same degree of attenuation of the same X-ray beam? Show your calculation. (2 marks)

(iii) Explain why X-ray radiographic imaging of the breast usually employs X-rays of lower energy (~20 keV) while examination of bony structures employs X-rays of energy around 100 keV. (2 marks)

(c) People are often concerned about the radiation exposure during medical examinations like X-ray radiographic imaging and computed tomography (CT) scans. Some information about the radiation dose is given below (Table d):

Source / item	Equivalent dose
Taking an X-ray radiographic image	0.1–0.2 mSv
A CT scan	1–10 mSv
Average weekly natural background dose for a person	about 0.05 mSv

Table d

(i) State one potential hazard of exposure to ionizing radiation on human bodies. (1 mark)

(ii) Explain why the equivalent dose of a CT scan is higher than that of taking an X-ray radiographic image. (2 marks)

(iii) Name one source contributing to the natural background dose. (1 mark)

Q27 Exam Report: In (a), most gave partially correct answers about the nature of radiations used. A few wrongly thought that a beta source rather than a gamma source was injected into the body. Many failed to state an X-ray tube as the source of X-rays. Most able candidates referred to the abnormal uptake of radioactive source by the target organ, which indicated a functional problem (e.g. hot spots in the case of bone scan). (b) was well answered.

27 Misc **HKDSE 2022 Paper 2 Q.4(b)**

Figure u is a radionuclide image showing the bone scan of a patient. Figure v shows a chest X-ray image of a person.



Fig u



Fig v

(a) In terms of **nature of radiation source and image production mechanism**, compare how the images in Figure u and Figure v are produced. No need to mention the instruments for detection and their detecting mechanisms. (3 marks)

(b) Radionuclide imaging is able to provide information that X-ray radiographic imaging cannot. Explain briefly. (2 marks)

Q26 Exam Report: Candidates' performance in (a) was satisfactory. A few did not realise that 'high-speed' electrons and a 'heavy' metal target were necessary for X-rays production. In (b)(i), most candidates were able to relate the higher density or larger effective atomic number of bone with the attenuation coefficients given. In (b)(ii), many knew how to apply the exponential equation to compute the answer, although some wrongly applied the ultrasound intensity reflection coefficient formula in this part. Part (b)(iii) was quite challenging. Many did not fully understand the physics principles involved in X-rays radiographic imaging, namely, how the penetrating power of X-rays depends on their energy and the relation between X-rays attenuation and contrast of image. The lower energy 20 keV X-rays, which is less penetrating, could be effectively attenuated by breast tissue during breast screening. Thus an image of transmitted X-rays with sufficient contrast to distinguish calcified tissues could be obtained. However, the contrast would be poor if highly penetrating 100 keV X-rays were used, as most X-rays penetrated the breast without much attenuation. Part (c)(i) was well answered. In (c)(ii), most were able to mention the multiple angles of exposure and longer exposure times for CT scans. Some misinterpreted the result of the CT scan (clear cross-section image) to account for the higher equivalent dose. Surprisingly, candidates' performance in (c)(iii) was poor. Many answers were not specific enough, taking 'sun, food, water, rock and building' as sources but without mentioning any of its radioactive substance content. Some claimed that electronic devices such as mobile phones were sources while a few confused the terms 'cosmic rays' and 'cosmic radiation'.

Know more

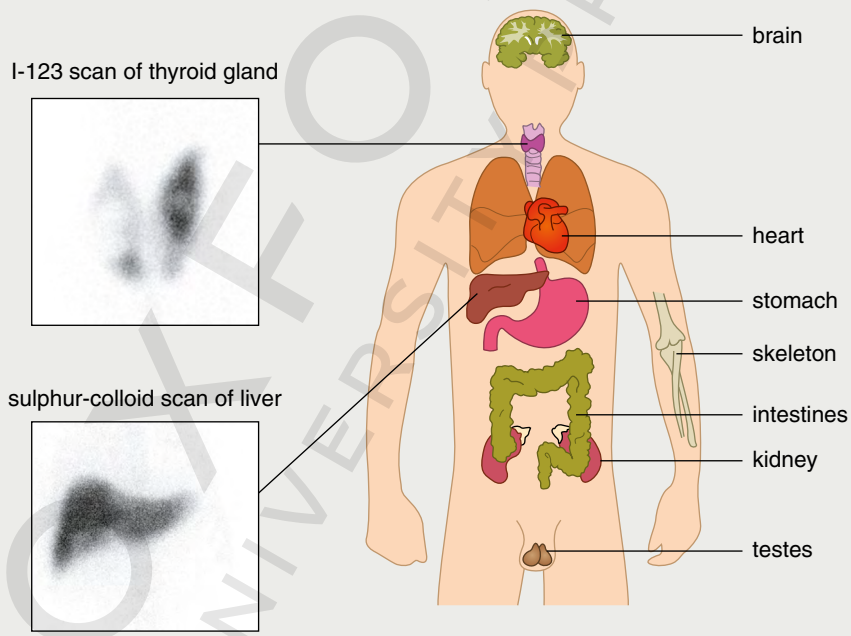
Examples of tracer absorption (p.151)

Natural absorption

The thyroid gland in the neck absorbs iodine obtained from food and converts them into thyroid hormone. If a radioisotope of iodine, I-123, is injected into the blood stream, it will be absorbed by the thyroid gland as well. The subsequent decay of the decay product of these trapped I-123 emits γ radiation that is detected by the camera to produce images of the thyroid gland.

Combined with substances

The liver acts as a cleaner for our body and will trap large particles (colloid) to metabolize them. If we combine a tracer with colloid, it will be taken up by the liver. One common way to do so is to combine radioactive technetium-99m with sulphur colloid to produce images of the liver.



Some body parts that radionuclide images can be taken.

Q24 Exam Report: Most candidates were able to identify the correct picture for radionuclide imaging in (a)(i) and mentioned that the patient needs taking in a radionuclide. However, many failed to point out that the image is formed by the accumulation of radionuclide at the target organ. They simply stated that an image is formed as there is radionuclide and ignored the crucial fact that its contrast is caused by the difference in concentration of radionuclide in the body. Some candidates mixed up the working mechanism with that of X-ray photography and wrongly reasoned by the attenuation through the body. In answering (a)(ii), many stated the advantages of other methods such as 'less harmful' and 'high resolution'. Part (b)(i) was well answered. Although most candidates knew the meaning of half-life, a lot of them failed to point out that the radionuclide is being removed by a biological process in (b)(ii). In (b)(iii), not many were able to get the correct effective half-life. Some of them simply used either physical or biological half-life in the calculation even though they knew how to apply the formula of exponential decay.

Q25 Exam Report: Most candidates in (a) knew how X-rays are produced. However, candidates' answers to parts (b) and (c)(i) involving CT scans were usually not concise. For the estimation of the equivalent background radiation dose in (c)(ii), most were able to obtain the correct answer. In (d), some candidates did not fully understand what the question asked for and they just worked out the intensity ratios separately instead of the overall ratio required. A few candidates failed to distinguish 'radiation' and 'ionizing radiation' in part (e).